On a Level-Set Characterization of the Value Function of an Integer Program and Its Application to Stochastic Programming

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Abstract: We propose a level-set approach to characterize the value function of a pure linear integer program with inequality constraints. We study theoretical properties of our characterization and show how they can be exploited to optimize a class of stochastic integer programs through a value function reformulation. Specifically, we develop algorithmic approaches that solve two-stage multidimensional knapsack problems with random budgets, yielding encouraging computational results.
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1 Introduction

The integer programming value function expresses the optimal objective value of an integer program (IP) as a function of its right-hand side. Given $A \in \mathbb{R}^{m\times n}$, $c \in \mathbb{R}^n$, and $\beta \in \mathbb{R}^m$, the value function z is defined as:

$$z(\beta) = \max\left\{c^\top x \mid x \in S(\beta)\right\},\ S(\beta) = \left\{x \in \mathbb{Z}^n_+ \mid Ax \le \beta\right\}. \tag{PIP}$$

We assume that all data are integral, i.e., $A \in \mathbb{Z}^{m\times n}$, $c \in \mathbb{Z}^n$, and $\beta \in \mathbb{Z}^m$. Thus, z maps $\mathbb{Z}^m$ to $\mathbb{Z}$.

Significant work concerning the theory of the IP value function include Blair and Jeroslow (1982) and Wolsey (1981). Additional relevant surveys on the topic include Güzelsoy and Ralphs (2007) and Nemhauser and Wolsey (1988). To the best of our knowledge, practical applications of the IP value function are rather limited, occurring mainly in the contexts of set partitioning (Klabjan 2004, 2007), lot sizing (Toriello and Nemhauser 2012), and stochastic integer programming (Ahmed et al. 2004, Kong et al. 2006, Özaltın et al. 2012, Schultz et al. 1998). It stands to reason that significant computational benefits for solving integer programs might be obtained given tractable characterizations of the value function.

In this paper we examine the value function of pure integer programs with inequality constraints. We show that the value function can be characterized entirely by its value on the set of vectors that are minimal with respect to their objective function level set (a level set of z is a set of the form $\{\beta \in \mathbb{Z}^m \mid z(\beta) = k\}$ for some $k \in \mathbb{Z}$). After showing that membership in this set is in general difficult to verify, we discuss a superset that maintains many of its desirable properties, and yet can be generated in a straightforward manner.

Our second, and potentially more significant, contribution is the usefulness of our characterization in the context of stochastic integer programming. Specifically, we consider two-stage pure integer programs with stochastic right-hand sides, which can be solved, among other alternatives, using a value function reformulation for both stages (Ahmed et al. 2004). As demonstrated in the literature (Kong et al. 2006, Özaltın et al. 2012), one of the main advantages of such approaches is their relative insensitivity to the number of variables and scenarios, though this comes at the price of sensitivity to both the number of constraints in each stage and the magnitude of right-hand sides because of value function storage requirements.

The modest number of constraints that some value function reformulation approaches can handle (no more than seven in Kong et al. (2006) and Özaltın et al. (2012)) imposes practical limitations. In this paper, we improve on such limitations by demonstrating that an optimal solution to the value function reformulation exists over the sets of level-set minimal vectors corresponding to both stages. These sets of level-set minimal vectors are generally much smaller than the entire set of interesting right-hand sides, leading to greatly reduced memory needs. Coupled with an efficient value function lookup for any given right-hand side, our approach is often able to effectively solve stochastic pure integer programs with a greater number of constraints. In our algorithmic developments and computational experiments we assume nonnegative constraint matrices for both stages, i.e., two-stage multidimensional knapsack problems with random budgets.

The remainder of this paper is organized as follows. In Section 2, we review properties of the IP value function and present our characterization using the concepts of level-set minimal vectors and integral monoids. Section 3 uses results from Section 2 to provide equivalent value function reformulations of the considered class of two-stage stochastic integer programs. We discuss algorithmic aspects in Section 4, including a global branch-and-bound approach that leverages properties of the IP value function in several ways. Section 5 describes the results of our computational experiments. For test instances amenable to solution via other approaches (e.g., Kong et al. (2006)), such methods should be preferred as they are typically more efficient; otherwise, our approach is able to solve instances with considerably more rows, possibly at the expense of time. We draw conclusions in Section 6, and include additional technical and algorithmic details in an e-companion.

2 Level-Set Characterization of the Value Function of a Pure Integer Program

2.1 Properties of the Value Function

Let a_j and c_j be the j^{th} column of the constraint matrix A and the j^{th} coefficient, respectively, of the objective function of (PIP). Given a right-hand side $\beta \in \mathbb{Z}^m$, define $\text{opt}(\beta) =$

$\{c^\top x \mid x \in S(\beta)\}$, and define $z(\beta) = -\infty$ if $S(\beta) = \emptyset$. We use 0 for a vector of zeros of conformable dimension.

Proposition 1 *Elementary properties of the value function of an integer program include:*

(a) $z(0) \in \{0, \infty\}$. *If* $z(0) = 0$, *then* $z(\beta) < \infty \ \forall \ \beta \in \mathbb{Z}^m$. *If* $z(0) = \infty$, *then* $z(\beta) = \pm\infty \ \forall \ \beta \in \mathbb{Z}^m$.

(b) $z(a_j) \geq c_j$ *for* $j = 1, \ldots, n$.

(c) z *is nondecreasing over* $\mathbb{Z}^m$.

(d) z *is superadditive over* $\mathbf{B} = \{\beta \in \mathbb{Z}^m \mid S(\beta) \neq \emptyset\}$. *That is, for all* $\beta_1, \beta_2 \in \mathbf{B}$, *if* $\beta_1 + \beta_2 \in \mathbf{B}$, *then* $z(\beta_1) + z(\beta_2) \leq z(\beta_1 + \beta_2)$.

Proposition 2 ***[Integer Complementary Slackness (ICS)]*** *If* $\hat{x} \in \mathrm{opt}(\beta)$ *for* $\beta \in \mathbb{Z}^m$, *then for all* $x \in \mathbb{Z}^n_+$ *such that* $x \leq \hat{x}$,

$$z(Ax) = c^\top x \quad \text{and} \quad z(Ax) + z(\beta - Ax) = z(Ax) + z(A(\hat{x} - x)) = z(\beta).$$

Corollary 1 ***[Column Elimination]*** *If* $z(a_j) > c_j$, *then* $\forall \ \beta \in \mathbb{Z}^m$ *and all* $\hat{x} \in \mathrm{opt}(\beta)$, $\hat{x}_j = 0$.

Proofs of the above results can be found in Wolsey (1981) and Nemhauser and Wolsey (1988). We next introduce an important set of right-hand side vectors that form the foundation of our study.

2.2 Level-Set Minimal Vectors

Definition 1 *A vector* $\beta \in \mathbb{Z}^m$ *is level-set minimal if* $z(\beta - e_i) < z(\beta)$ *for all* $i = 1, \ldots, m$, *where* e_i *is the* i^{th} *unit vector. Let* $\bar{\mathbf{B}} \subseteq \mathbb{Z}^m$ *be the set of all level-set minimal vectors.*

Level-set minimal vectors generalize the notion of *minimal tenders* introduced by Kong et al. (2006) in the context of a two-stage stochastic IP assuming a nonnegative constraint matrix. Our definition requires no restrictions on the sign of data through two modest assumptions, which we assume hold throughout the remainder of the paper:

A1. $z(0) = 0$.

A2. $\bar{\mathbf{B}} \neq \emptyset$.

The following results specifically motivate these assumptions, identifying necessary and sufficient conditions under which they hold. For the sake of brevity, some proofs have been omitted.

Lemma 1 $\bar{\mathbf{B}} \subseteq \mathbf{B}$, *and* $\forall \ \beta \in \mathbf{B} \backslash \bar{\mathbf{B}}$, $\exists \ i \in \{1, \ldots, m\}$ *such that* $S(\beta - e_i) \neq \emptyset$ *and* $z(\beta - e_i) = z(\beta)$.

Lemma 2 $z(0) = 0$ *if and only if* $\nexists \ \tilde{x} \in \mathbb{Z}^n_+$ *such that* $A\tilde{x} \leq 0$ *with* $c^\top \tilde{x} > 0$.

Lemma 3 *(Gordan 1873). Every sequence* $\{x^{(1)}, x^{(2)}, \ldots\} \subseteq \mathbb{Z}_+^n$ *with* $x^{(i)} \not\leq x^{(j)}$ *whenever* $i < j$*, is finite.*

Lemma 4 *Let* $z(0) = 0$*. Then* $\bar{\mathbf{B}} = \emptyset$ *if and only if* $\exists\, \tilde{x} \in \mathbb{Z}_+^n$ *such that* $A\tilde{x} \leq 0$*,* $A\tilde{x} \neq 0$ *with* $c^\top \tilde{x} = 0$*.*

Proof. We first note that for any $\beta \in \mathbf{B}$, $z(0) = 0$ and Proposition (a) imply that $\text{opt}(\beta) \neq \emptyset$.

$\Leftarrow$ Suppose $\exists\, \tilde{x} \in \mathbb{Z}_+^n$ such that $A\tilde{x} \leq 0$, $A\tilde{x} \neq 0$ with $c^\top \tilde{x} = 0$. Consider any $\beta \in \mathbf{B}$. Let $\hat{x} \in \text{opt}(\beta)$. Then $(\hat{x} + \tilde{x}) \in \text{opt}(\beta)$ since $A(\hat{x} + \tilde{x}) \leq A\hat{x} \leq \beta$ and $c^\top(\hat{x} + \tilde{x}) = z(\beta)$. Note $A(\hat{x}+\tilde{x}) \neq A\hat{x}$. Thus $\exists\, i \in \{1, \ldots, m\}$ such that $(A(\hat{x}+\tilde{x}))_i < \beta_i$ and yet $z(A(\hat{x}+\tilde{x})) = z(\beta)$, implying $\beta \notin \bar{\mathbf{B}}$. Thus, $\bar{\mathbf{B}} = \emptyset$.

$\Rightarrow$ Suppose $\bar{\mathbf{B}} = \emptyset$, but yet $\nexists\, \tilde{x} \in \mathbb{Z}_+^n$ such that $A\tilde{x} \leq 0$, $A\tilde{x} \neq 0$, and $c^\top \tilde{x} = 0$. For any $\beta \in \mathbf{B}$, choose some $x^{(0)}$ such that $x^{(0)} \in \text{opt}(\beta)$, so that $Ax^{(0)} \leq \beta$, $z(Ax^{(0)}) = z(\beta)$, and $\text{opt}(Ax^{(0)}) \subseteq \text{opt}(\beta)$ by Proposition (c). By hypothesis, $Ax^{(0)} \notin \bar{\mathbf{B}}$, so by Lemma 1, $\exists\, i' \in \{1, \ldots, m\}$ such that $S(Ax^{(0)} - e_{i'}) \neq \emptyset$ and $z(Ax^{(0)} - e_{i'}) = z(Ax^{(0)}) = z(\beta)$. Note that $x^{(0)} \in \text{opt}(Ax^{(0)})$ and $x^{(0)} \notin S(Ax^{(0)} - e_{i'})$ imply $\text{opt}(Ax^{(0)} - e_{i'}) \subset \text{opt}(Ax^{(0)}) \subseteq \text{opt}(\beta)$.

For the case of $j > 0$, by hypothesis $Ax^{(j-1)} \notin \bar{\mathbf{B}}$, so by Lemma 1 $\exists\, i'' \in \{1, \ldots, m\}$ such that $S(Ax^{(j-1)} - e_{i''}) \neq \emptyset$ and $z(Ax^{(j-1)} - e_{i''}) = z(Ax^{(j-1)})$. Choose $x^{(j)} \in \text{opt}(Ax^{(j-1)} - e_{i''})$, so that $Ax^{(j)} \leq Ax^{(j-1)} - e_{i''}$, $z(Ax^{(j)}) = z(Ax^{(j-1)} - e_{i''})$, and $\text{opt}(Ax^{(j)}) \subseteq \text{opt}(Ax^{(j-1)} - e_{i''}) \subset \text{opt}(Ax^{(j-1)}) \subseteq \text{opt}(\beta)$. This process generates an infinite sequence of vectors $x^{(j)} \in \mathbb{Z}_+^n$, $j \in \mathbb{Z}_+$ where for $j < k$, $Ax^{(j)} \geq Ax^{(k)}$, $Ax^{(j)} \neq Ax^{(k)}$, $\text{opt}(Ax^{(j)}) \supset \text{opt}(Ax^{(k)})$, and $z(Ax^{(j)}) = z(Ax^{(k)})$. Because $x \in \mathbb{Z}_+^n$, by Lemma 3 there exist j', k' with $j' < k'$ such that $x^{(j')} \leq x^{(k')}$, $x^{(j')} \neq x^{(k')}$. Consider vector $\tilde{x} = x^{(k')} - x^{(j')}$, for which $\tilde{x} \in \mathbb{Z}_+^n$, $\tilde{x} \neq 0$, and $A\tilde{x} \leq 0$, $A\tilde{x} \neq 0$. Because $z(Ax^{(j')}) = z(Ax^{(k')})$, it follows that $c^\top \tilde{x} = 0$, a contradiction. ∎

This result implies that under Assumptions **A1** and **A2** for any $x \in \mathbb{Z}_+^n$ such that $Ax \leq 0$ and $Ax \neq 0$, $c^\top x < 0$. Moreover, under our assumptions, for any right-hand side $\beta \in \mathbf{B}$, there exists a right-hand side that is not larger than β, provides the same objective function value, and is also a level-set minimal vector. We formalize this observation through the following statements.

Proposition 3 *For any* $\beta \in \mathbf{B}$ *there exists* $\tilde{\beta} \in \bar{\mathbf{B}}$ *such that* $\tilde{\beta} \leq \beta$ *and* $z(\tilde{\beta}) = z(\beta)$*.*

Theorem 1 *For any* $\beta \in \mathbb{Z}^m$*,*

$$z(\beta) = \begin{cases} \max\limits_{\tilde{\beta} \in \bar{\mathbf{B}}:\ \tilde{\beta} \leq \beta} z(\tilde{\beta}) & \text{if } \beta \in \mathbf{B}; \\ -\infty & \text{if } \beta \notin \mathbf{B}. \end{cases} \tag{1}$$

Note that Theorem 1 extends to $\beta \in \mathbb{R}^m$ by Proposition (c) and our assumption that $A \in \mathbb{Z}^{m \times n}$.

Proposition 4 *For any* $\beta \in \bar{\mathbf{B}}$ *and any* $\hat{x} \in \text{opt}(\beta)$*,* $A\hat{x} = \beta$*.*

Corollary 2 *For any* $\beta \in \mathbf{B}$*, if* $\exists\, \hat{x} \in \text{opt}(\beta)$ *such that* $A\hat{x} \neq \beta$*, then* $\beta \notin \bar{\mathbf{B}}$*.*

The following result demonstrates downward inclusion in $\bar{\mathbf{B}}$ for those Ax transformations of nonnegative x vectors dominated by any $\hat{x} \in \text{opt}(A\hat{x})$ for which $A\hat{x}$ is level-set minimal.

Proposition 5 *If $\hat{x} \in \text{opt}(A\hat{x})$ and $A\hat{x} \in \bar{\mathbf{B}}$, then $\forall\, x \in \mathbb{Z}_+^n$ such that $x \leq \hat{x}$, $Ax \in \bar{\mathbf{B}}$.*

Proof. Suppose for some $\beta \in \mathbf{B}$ and $\hat{x} \in \text{opt}(\beta)$ with $A\hat{x} \in \bar{\mathbf{B}}$, there is some $\bar{x} \in \mathbb{Z}_+^n$ such that $\bar{x} \leq \hat{x}$, but yet $A\bar{x} \notin \bar{\mathbf{B}}$. By Lemma 1 there exists $i \in \{1, \ldots, m\}$ such that $S(A\bar{x} - e_i) \neq \emptyset$ and $z(A\bar{x} - e_i) = z(A\bar{x}) = c^\top \bar{x}$ (by Proposition 2). Let $\breve{x} \in \text{opt}(A\bar{x} - e_i)$ and thus $\breve{x} \in S(A\bar{x} - e_i)$ with $c^\top \breve{x} = c^\top \bar{x}$.

Consider the nonnegative integer vector $\tilde{x} = \hat{x} - \bar{x} + \breve{x}$. Then $A\tilde{x} = A(\hat{x} - \bar{x} + \breve{x}) \leq A\hat{x} - A\bar{x} + A\bar{x} - e_i = A\hat{x} - e_i$, and so $\tilde{x} \in S(A\hat{x} - e_i)$. Similarly, $c^\top \tilde{x} = c^\top(\hat{x} - \bar{x} + \breve{x}) = c^\top \hat{x} - c^\top \bar{x} + c^\top \breve{x} = c^\top \hat{x}$, implying $z(A\hat{x} - e_i) \geq z(A\hat{x})$, contradicting the hypothesis that $A\hat{x} \in \bar{\mathbf{B}}$. ∎

Corollary 3 *Let $\hat{x} \in \text{opt}(A\hat{x})$ and $A\hat{x} \in \bar{\mathbf{B}}$. For any $1 \leq j \leq n$, if $\hat{x}_j \geq 1$ then for all integer k such that $1 \leq k \leq \hat{x}_j$, we have $ka_j \in \bar{\mathbf{B}}$.*

Proposition 5 and Corollary 3 generalize Lemma 4 of Kong et al. (2006), who show the case of $k = 1$ for $A \in \mathbb{Z}_+^{m \times n}$, i.e., if $\hat{x}_j \geq 1$ then $a_j \in \bar{\mathbf{B}}$. We can reduce the size of A in (PIP) by eliminating columns. Let $\mathcal{A}$ be the index set of columns of A that are level-set minimal. For any $\beta \in \mathbb{Z}^m$, define:

$$z'(\beta) = \max \left\{ \sum_{j \in \mathcal{A}} c_j x_j \;\middle|\; x \in S'(\beta) \right\},\; S'(\beta) = \left\{ x \in \mathbb{Z}_+^n \;\middle|\; \sum_{j \in \mathcal{A}} a_j x_j \leq \beta \right\}. \tag{2}$$

Proposition 6 *For any $\beta \in \mathbf{B}$, $z'(\beta) = z(\beta)$.*

Proposition 6 follows from arguments in Kong et al. (2006) with modifications to accommodate level-set minimal vectors. It gives an alternate column elimination method as compared with Corollary 1. Neither approach dominates the other, as demonstrated in the following example.

Consider the following value function:

$$\begin{aligned} z([\beta_1, \beta_2]^\top) = \max_{x \in \mathbb{Z}_+^4}\; & 3x_1 + 2x_2 + 4x_3 + 12x_4 \\ \text{subject to} \quad & -1x_1 + 2x_2 + 1x_3 + 5x_4 \leq \beta_1 \\ & 3x_1 - 1x_2 + 2x_3 + 3x_4 \leq \beta_2. \end{aligned}$$

The constraint matrix column corresponding to x_3 is by definition level-set minimal, but can be eliminated via Corollary 1, because $z([1, 2]^\top) = 5 > 4$. Though the constraint matrix column corresponding to x_4 has $z([5, 3]^\top) = 12$ (and so will not be eliminated by Corollary 1), it is not level-set minimal, and so can be eliminated. We make use of these column elimination techniques in Sections 3.3 and 4. Some additional properties of $\bar{\mathbf{B}}$ are provided below.

Proposition 7 *For any $\beta \in \mathbf{B}$, let $\hat{x} \in \mathrm{opt}(\beta)$.*

(a) If $\nexists\ \tilde{x} \in \mathrm{opt}(\beta)$ where $A\tilde{x} \leq A\hat{x}$ and $A\tilde{x} \neq A\hat{x}$, then for all $x \in \mathbb{Z}^n_+$ such that $x \leq \hat{x}$, $Ax \in \bar{\mathbf{B}}$.

(b) If $\mathrm{opt}(\beta) = \{\hat{x}\}$, then for all $x \in \mathbb{Z}^n_+$ such that $x \leq \hat{x}$, $Ax \in \bar{\mathbf{B}}$.

(c) If $\beta \in \bar{\mathbf{B}}$ and A has full column rank, then $\hat{x} \in \mathrm{opt}(\beta)$ is the unique optimal solution for $z(\beta)$.

Proof. (a) Observe that if $\nexists\ \tilde{x} \in \mathrm{opt}(\beta)$, where $A\tilde{x} \leq A\hat{x}$ and $A\tilde{x} \neq A\hat{x}$, then $A\hat{x} \in \bar{\mathbf{B}}$. The result then follows directly from Proposition 5. We omit the similar proofs of (b) and (c). ∎

It may be computationally advantageous to optimize over $\bar{\mathbf{B}}$, particularly if $|\bar{\mathbf{B}}| << |\mathbf{B}|$. However, we note that there exist cases where $\bar{\mathbf{B}} = \mathbf{B}$, which we consider in the following two results.

Proposition 8 *If $\forall\ i = 1, \ldots, m\ \exists\ x^{(i)} \in \mathbb{Z}^n_+$ such that $Ax^{(i)} = e_i$ with $c^\top x^{(i)} > 0$, then $\bar{\mathbf{B}} = \mathbf{B}$.*

Proof. Suppose $\forall\ i \in \{1, \ldots, m\}$ there exists $x^{(i)} \in \mathbb{Z}^n_+$ such that $Ax^{(i)} = e_i$ with $c^\top x^{(i)} > 0$, but yet $\bar{\mathbf{B}} \neq \mathbf{B}$. Because $\bar{\mathbf{B}} \subseteq \mathbf{B}$ by Lemma 1, this implies the existence of $\beta \in \mathbf{B}\backslash\bar{\mathbf{B}}$. Then by Lemma 1 $\exists\ i \in \{1, \ldots, m\}$ such that $S(\beta - e_i) \neq \emptyset$ and $z(\beta - e_i) = z(\beta)$. Choose $\hat{x} \in \mathrm{opt}(\beta - e_i) \subseteq \mathrm{opt}(\beta)$. Then $A(\hat{x} + x^{(i)}) = A\hat{x} + Ax^{(i)} \leq \beta - e_i + e_i = \beta$, implying $(\hat{x} + x^{(i)}) \in S(\beta)$. Yet $c^\top(\hat{x} + x^{(i)}) = c^\top\hat{x} + c^\top x^{(i)} > c^\top\hat{x}$, contradicting the hypothesis that $\hat{x} \in \mathrm{opt}(\beta)$. ∎

Proposition 9 *Let $A \in \mathbb{Z}^{m\times n}_+$ and $c > 0$. Then $\bar{\mathbf{B}} = \mathbf{B}$ if and only if A contains the identity submatrix $I^{m\times m}$.*

Proof. $\Rightarrow$ Suppose $\bar{\mathbf{B}} = \mathbf{B}$ but yet A does not contain $I^{m\times m}$. Then there exists $e_i \in \mathbb{Z}^m$ for which $a_j \neq e_i\ \forall\ j \in \{1, \ldots, n\}$. $0 \in S(e_i)$ implies $e_i \in \mathbf{B}$. By the integrality and nonnegativity of A, $\nexists\ x \in \mathbb{Z}^n_+$ such that $Ax = e_i$, and so $e_i \notin \bar{\mathbf{B}}$ by Proposition 4. Hence $\bar{\mathbf{B}} \neq \mathbf{B}$, a contradiction.

$\Leftarrow$ See Kong et al. (2006). ∎

While, by Theorem 1, the value function can be completely characterized by its values at the members of $\bar{\mathbf{B}}$, inclusion in $\bar{\mathbf{B}}$ is unfortunately difficult to verify.

Theorem 2 *Given $\beta \in \mathbf{B}$ and the value of $z(\beta)$, the problem of checking if $\beta \notin \bar{\mathbf{B}}$ is NP-complete.*

Proof. The problem is in NP since given the value of $z(\beta)$ the fact that $\beta \notin \bar{\mathbf{B}}$ can be verified in polynomial time if we know $\tilde{x} \in \mathbb{Z}^n_+$ such that $A\tilde{x} \leq \beta$, $A\tilde{x} \neq \beta$ and $z(A\tilde{x}) = z(\beta)$ (Proposition 7).

Consider the integer knapsack feasibility problem, which is NP-complete (Papadimitriou and Steiglitz 1998). Given nonnegative integers $s_1 < s_2 < \ldots < s_n$, and K, we need to check whether there exist nonnegative integers $x_1, x_2, \ldots, x_n$ satisfying

$$s_1x_1 + s_2x_2 + \ldots + s_nx_n = K. \tag{3}$$

Given an instance of this problem consider the following parameterized pure integer program:

$$z(\beta) = \max_{x \in \mathbb{Z}_+^{n+1}} \sum_{i=1}^{n} s_i x_i \; + \; K x_{n+1} \tag{4}$$

$$\text{subject to} \qquad 2\sum_{i=1}^{n} s_i x_i + (2K+1)x_{n+1} \leq \beta. \tag{5}$$

For $\beta = 2K+1$ vector $\hat{x} = (0,0,\ldots,0,1) \in \mathrm{opt}(\beta)$ and $z(\beta) = K$. To check whether $\beta = 2K+1$ is not level-set minimal, it is sufficient to verify whether $z(2K) = K$. Observe that for $\beta = 2K$ any feasible solution of (4) – (5) must have $x_{n+1} = 0$, which immediately implies that $z(2K) = K$ if and only if there exists nonnegative integers x_1, …, x_n such that (3) is satisfied.

■

2.3 Integral Monoids

Observe that Theorem 2 holds even when $\hat{x} \in \mathrm{opt}(\beta)$ is known and $m = 1$, indicating the difficulty of verifying whether β is level-set minimal. This fact motivates our discussion of a superset of level-set minimal vectors that can be constructed in a straightforward manner, and yet maintains many of the desirable properties.

Because of our further algorithmic and computational study in the context of stochastic integer programming, in this section we generalize the definition of $\mathbf{B}$ as follows. Let $\mathbf{B}' \subseteq \{\beta \in \mathbb{Z}^m \mid S(\beta) \neq \emptyset\ \}$, i.e., $\mathbf{B}'$ is some subset of right-hand sides that induce a nonempty feasible region. Redefine $\mathbf{B} = \mathbf{B}' \cup \{\tilde{\beta} \in \mathbb{Z}^m \mid \tilde{\beta}$ is level-set minimal and $\exists \beta \in \mathbf{B}'$ such that $\tilde{\beta} \leq \beta$ and $z(\tilde{\beta}) = z(\beta)\}$, i.e., $\forall\ \beta \in \mathbf{B}$ if $\tilde{\beta} \in \mathbb{Z}^m$ is level-set minimal, $\tilde{\beta} \leq \beta$ and $z(\tilde{\beta}) = z(\beta)$, then $\tilde{\beta} \in \mathbf{B}$. Such a construction of $\mathbf{B}$ is possible due to Proposition 3. Let $\bar{\mathbf{B}} \subseteq \mathbf{B}$ be the set of all level-set minimal vectors in $\mathbf{B}$.

An *integral monoid* (Jeroslow 1978) is a set of vectors in $\mathbb{Z}^m$ that forms a semi-group under addition. Let M be the integral monoid generated by nonnegative integer linear combinations of the columns of A that are level-set minimal, and define the *truncated integral monoid* as $\bar{M} = M \cap \mathbf{B}$. The next result follows from Proposition 4.

Proposition 10 *The set of level-set minimal vectors $\bar{\mathbf{B}} \subseteq \bar{M}$.*

Definition 2 *The partial value function $\bar{z}$ of a general integer program is the value function z defined over the restricted domain of the truncated integral monoid $\bar{M}$, that is,*

$$\bar{z}(\pi) = \max\left\{c^\top x \mid x \in S(\pi)\right\}, \text{ where } S(\pi) = \left\{x \in \mathbb{Z}_+^n \mid Ax \leq \pi\right\} \text{ for } \pi \in \bar{M}.$$

For the remainder of the text we reserve our use of the term *value function* as per the classical definition in (PIP). Note that over $\bar{M}$, the partial value function $\bar{z}$ precisely coincides with the value function z, and moreover the results of Proposition 1 also hold for $\bar{z}$ (with domains adjusted where necessary, i.e., $\beta \in \bar{M} \subseteq \mathbf{B}$).

Theorem 3 *For any $\beta \in \mathbf{B}$,*

$$z(\beta) = \max_{\pi \in \bar{M} : \pi \leq \beta} \bar{z}(\pi). \tag{6}$$

3 Level-Set Approach for Solving a Class of Stochastic Pure Integer Programs

We next illustrate that the value function characterization introduced in Section 2 can be exploited to solve the following class of stochastic pure integer programs:

$$\begin{aligned} \max_{x \in \mathbb{Z}_+^{n_1}} \quad & c^\top x + \mathbb{E}_{\omega \in \Omega} Q(x, \omega) \\ \text{subject to} \quad & Ax \le b, \end{aligned} \tag{P1}$$

where, given $x \in \mathbb{Z}_+^{n_1}$ and $\omega \in \Omega$, the second-stage recourse function $Q(x, \omega)$ is defined as:

$$\begin{aligned} Q(x, \omega) \;=\; & \max_{y \in \mathbb{Z}_+^{n_2}} \; d^\top y \\ \text{subject to} \quad & Wy \le h(\omega) \; - \; Tx. \end{aligned}$$

We use random variable ω from probability space $(\Omega, \mathcal{F}, \mathcal{P})$ to characterize the uncertain parameter realizations (scenarios). The total number of constraints and decision variables is m_i and n_i, respectively, for stages $i = 1, 2$. Known entities include constraint matrix A and vectors c and b. We assume the technology matrix T, the recourse matrix W and the second-stage objective d are deterministic, leaving $h(\cdot)$ as the lone stochastic component of the problem. We also make the following additional assumptions for (P1):

A3. The random variable ω is discretely distributed and has finite support.

A4. The first-stage feasibility set $X = \left\{ x \mid x \in \mathbb{Z}_+^{n_1},\; Ax \le b \right\}$ is nonempty and bounded.

A5. $Q(x, \omega)$ is finite for all $x \in X$ and $\omega \in \Omega$.

A6. The first-stage constraint matrix A, technology matrix T and recourse matrix W are nonnegative and integral, i.e., $A \in \mathbb{Z}_+^{m_1 \times n_1}$, $T \in \mathbb{Z}_+^{m_2 \times n_1}$ and $W \in \mathbb{Z}_+^{m_2 \times n_2}$.

The validity of Assumption **A3** is due to Schultz (1995), who shows that the optimal solution to a stochastic program with continuously distributed ω can be approximated via a discrete distribution to any desired accuracy. The finiteness of X is assured by Assumption **A4**. Assumption **A5** establishes relatively complete recourse, that is, $Q(x, \omega)$ has a feasible integer completion for all $x \in X$ and $\omega \in \Omega$. Assumptions similar to **A3** – **A5** are common in many of the related works in the literature, e.g. Ahmed et al. (2004), Carøe and Tind (1998), Kong et al. (2006), Özaltın et al. (2012), Schultz et al. (1998), Sen and Higle (2005), and Ntaimo (2010). While the nonnegativity restriction in Assumption **A6** is not necessary for equivalent value function reformulations in this section, we use this restriction in subsequent Sections 4 and 5 for our algorithmic developments and computational experiments. The nonnegativity restriction is also present in computational experiments of other closely related works such as Kong et al. (2006) and Özaltın et al. (2012). Based on Assumption **A6**, without loss of generality we assume that $c \in \mathbb{Z}_+^{n_1}$, $d \in \mathbb{Z}_+^{n_2}$, $b \in \mathbb{Z}_+^{m_1}$ and $h(\omega) \in \mathbb{Z}_+^{m_2} \; \forall \; \omega \in \Omega$. Therefore, the class of stochastic integer programs we consider can be viewed as two-stage multidimensional knapsack problems with random budgets.

For standard texts on the theory, algorithms and applications of stochastic programming we refer to Birge and Louveaux (2011), Shapiro et al. (2009), and Ruszczyński and Shapiro (2003).

3.1 Value Function Reformulation

We reformulate (P1) with a value function transformation of the variable space, and subsequently apply results from Section 2 to obtain an optimal solution. Without loss of generality, we incorporate the set of first-stage constraints $Ax \leq b$ into the second stage by augmentation, so that T becomes $\binom{A}{T}$, W becomes $\binom{0}{W}$, and $h(\omega)$ becomes $\binom{b}{h(\omega)}$. We define $m = m_1 + m_2$.

For any $\beta_1 \in \mathbb{Z}^m$, define the first-stage value function as:

$$\psi(\beta_1) = \max\left\{c^\top x \mid x \in S_1(\beta_1)\right\}, \; S_1(\beta_1) = \left\{x \in \mathbb{Z}_+^{n_1} \mid Tx \leq \beta_1\right\}. \tag{7}$$

For any $\beta_2 \in \mathbb{Z}^m$, define the second-stage value function as:

$$\phi(\beta_2) = \max\left\{d^\top y \mid y \in S_2(\beta_2)\right\}, \quad S_2(\beta_2) = \left\{y \in \mathbb{Z}_+^{n_2} \mid Wy \leq \beta_2\right\}. \tag{8}$$

Define $\mathbf{B}^1 = \{\beta_1 \in \mathbb{Z}^m \mid S_1(\beta_1) \neq \emptyset \text{ and } \forall\, \omega \in \Omega,\; S_2(h(\omega) - \beta_1) \neq \emptyset\}$, that is, $\mathbf{B}^1$ is the set of vectors $\beta_1 \in \mathbb{Z}^m$ such that there exists a feasible first-stage $x \in S_1(\beta_1)$, and has a feasible integer completion for all $\omega \in \Omega$. Define $\mathbf{B}^2 = \{\beta_2 \in \mathbb{Z}^m \mid S_2(\beta_2) \neq \emptyset \text{ and } \exists\, \beta_1 \in \mathbf{B}^1,\; \exists\, \omega \in \Omega, \text{ such that } \beta_2 \leq h(w) - \beta_1\}$. Note that **A4** and **A5** imply $\mathbf{B}^1 \neq \emptyset$ and $\mathbf{B}^2 \neq \emptyset$. Also, $\mathbf{B}^2 \supseteq \cup_{\beta_1 \in \mathbf{B}^1} \cup_{\omega \in \Omega} \{h(\omega) - \beta_1\}$, i.e., $\mathbf{B}^2$ contains all vectors $\beta_2 \in \mathbb{Z}^m$ such that there exist $\beta_1 \in \mathbf{B}^1$ and $\omega \in \Omega$ with $\beta_2 = h(\omega) - \beta_1$. We then reformulate (P1) as:

$$\max_{\beta_1 \in \mathbf{B}^1} \left\{\, \psi(\beta_1) + \mathbb{E}\phi(h(\omega) - \beta_1)\right\}. \tag{P2}$$

Variables β_1 in (P2) are known as *tender variables* in the stochastic programming literature.

Theorem 4 *(Ahmed et al. 2004, Kong et al. 2006). Let β_1^* solve* (P2). *Then an optimal solution to* (P1) *is $\hat{x} \in \left\{c^\top x \mid Tx \leq \beta_1^*,\; x \in \mathbb{Z}_+^{n_1}\right\}$. Moreover, the optimal objective values are equal.*

3.2 Optimizing over Level-Set Minimal Vectors

Let $\bar{\mathbf{B}}^k$ be the set of level-set minimal vectors in $\mathbf{B}^k$, $k = 1, 2$.

Theorem 5 *There exists an optimal solution to* (P2) *that is a level-set minimal vector. That is,*

$$\max_{\beta_1 \in \bar{\mathbf{B}}^1} \{\psi(\beta_1) \;+\; \mathbb{E}\phi(h(\omega) - \beta_1)\} \;=\; \max_{\beta_1 \in \mathbf{B}^1} \{\psi(\beta_1) \;+\; \mathbb{E}\phi(h(\omega) - \beta_1)\}.$$

Proof. Under **A1** and **A2**, the existence of such a level-set minimal vector is given by Proposition 3. The remainder of the proof follows from Theorem 6 of Kong et al. (2006). ∎

We next demonstrate that an optimal solution to (P2) can be identified using only level-set minimal vector information from $\bar{\mathbf{B}}^1$ and $\bar{\mathbf{B}}^2$.

Lemma 5 *For any pair (β_1, ω) with $\beta_1 \in \bar{\mathbf{B}}^1$ and $\omega \in \Omega$, there exists $\beta_2^\omega \in \bar{\mathbf{B}}^2$ with $\beta_2^\omega \leq h(\omega) - \beta_1$ and $\phi(\beta_2^\omega) = \phi(h(\omega) - \beta_1)$.*

Proof. Let $\delta^\omega = h(\omega) - \beta_1$. If $\delta^\omega \in \bar{\mathbf{B}}^2$, the result follows immediately. If $\delta^\omega \notin \bar{\mathbf{B}}^2$, then by Lemma 1 $\exists\ i \in \{1, \dots, m\}$ such that $S_2(\delta^\omega - e_i) \neq \emptyset$ and $\phi(\delta^\omega - e_i) = \phi(\delta^\omega)$. If $(\delta^\omega - e_i) \in \bar{\mathbf{B}}^2$, the result follows. Otherwise $(\delta^\omega - e_i) \notin \bar{\mathbf{B}}^2$, and by repeating this argument, by Proposition 3 we eventually reach $\beta_2^\omega \in \bar{\mathbf{B}}^2$ for which $\beta_2^\omega \leq \delta^\omega$ with $\phi(\beta_2^\omega) = \phi(\delta^\omega)$. ∎

The following reformulation considers only level-set minimal vectors from $\bar{\mathbf{B}}^1$ and $\bar{\mathbf{B}}^2$:

$$\max_{\beta_1 \in \bar{\mathbf{B}}^1, \beta_2^\omega \in \bar{\mathbf{B}}^2} \left\{ \psi(\beta_1) + \mathbb{E}\phi(\beta_2^\omega) \mid \beta_2^\omega \leq h(\omega) - \beta_1,\ \forall\ \omega \in \Omega \right\}. \tag{9}$$

Theorem 6 *Any optimal solution $\beta_1 \in \bar{\mathbf{B}}^1$ to (9) is an optimal solution to (P2). That is,*

$$\max_{\beta_1 \in \bar{\mathbf{B}}^1, \beta_2^\omega \in \bar{\mathbf{B}}^2} \{\psi(\beta_1) + \mathbb{E}\phi(\beta_2^\omega) \mid \beta_2^\omega \leq h(\omega) - \beta_1,\ \forall\ \omega \in \Omega\} = \max_{\beta_1 \in \mathbf{B}^1} \{\psi(\beta_1) + \mathbb{E}\phi(h(\omega) - \beta_1)\}.$$

Proof. Theorem 1 demonstrates the first-stage value function equivalence, and for any $\beta_1 \in \bar{\mathbf{B}}^1$, Lemma 5 proves the existence of $\beta_2^\omega \in \bar{\mathbf{B}}^2$ with $\beta_2^\omega \leq h(\omega) - \beta_1$ such that $\phi(\beta_2^\omega) = \phi(h(\omega) - \beta_1)$. ∎

3.3 Optimizing over Integral Monoids

Let set $\mathcal{T}$ index the columns of T that are level-set minimal, and let $\mathcal{W}$ be defined in similar fashion. Define M^1 to be the integral monoid generated by the nonnegative integer linear combinations of the columns $\{t_j\}_{j \in \mathcal{T}}$, and likewise, let M^2 be generated from the columns $\{w_j\}_{j \in \mathcal{W}}$. For $k = 1, 2$, define $\bar{M}^k = M^k \cap \mathbf{B}^k$ to be the corresponding truncated integral monoids, that is, those monoid elements contained within $\mathbf{B}^k$. We use π to denote individual first-stage monoid elements and τ for second-stage elements. Finally, define the partial value functions $\bar{\psi}$ and $\bar{\phi}$ for the first and second stages, respectively, and in view of (P2) and Theorem 6, consider the following problem:

$$\max_{\pi \in \bar{M}^1, \tau^\omega \in \bar{M}^2} \left\{ \bar{\psi}(\pi) + \mathbb{E}\bar{\phi}(\tau^\omega) \mid \tau^\omega \leq h(\omega) - \pi,\ \forall\ \omega \in \Omega \right\}. \tag{P3}$$

Theorem 7 *Any optimal solution $\pi^* \in \bar{M}^1$ to (P3) is an optimal solution to (P2), i.e.,*

$$\max_{\pi \in \bar{M}^1, \tau^\omega \in \bar{M}^2} \left\{ \bar{\psi}(\pi) + \mathbb{E}\bar{\phi}(\tau^\omega) \mid \tau^\omega \leq h(\omega) - \pi,\ \forall\ \omega \in \Omega \right\} = \max_{\beta_1 \in \mathbf{B}^1} \{\psi(\beta_1) + \mathbb{E}\phi(h(\omega) - \beta_1)\}.$$

Proof. Theorem 3 gives the first-stage value function equivalence, and because $\bar{\mathbf{B}}^2 \subseteq \bar{M}^2$, for any $\pi \in \bar{M}^1$, Lemma 5 proves the existence of $\tau^\omega \in \bar{M}^2$ with $\tau^\omega \leq h(\omega) - \pi$ and $\phi(\tau^\omega) = \phi(h(\omega) - \pi)$. ∎

Theorem 7 demonstrates that an optimal solution to (P2) exists over $\bar{M}^1$. This result can be further strengthened by applying Corollary 1 to reduce the sizes of $\mathcal{T}$ and $\mathcal{W}$ prior to generating their respective integral monoids. Let $\overline{T}$ and $\overline{\mathcal{T}}$ be the reduced constraint matrix and index set, respectively, i.e., $\overline{\mathcal{T}}$ indexes those columns that are both level-set minimal and satisfy $\psi(t_j) = c_j$. Let $\overline{W}$ and $\overline{\mathcal{W}}$ be defined in similar fashion. By only generating $\bar{M}^1$ and $\bar{M}^2$ over these columns, it can be shown that Theorem 7 still holds. We carry this out in Section 4, relegating most of the technical and implementation details to Section 7 of the e-companion.

			First Stage			Second Stage			Stochastic Components			
Author / Year	MS	CS	$\mathbb{R}$	$\mathbb{B}$	$\mathbb{Z}$	$\mathbb{R}$	$\mathbb{B}$	$\mathbb{Z}$	$T(\omega)$	$W(\omega)$	$d(\omega)$	$h(\omega)$
Carøe and Tind (1998)			•					•	•			•
Schultz et al. (1998)		•	•					•				•
Sherali and Fraticelli (2002)				•		•	•		•	•	•	•
Alonso-Ayuso et al. (2003)	•	•		•		•	•				•	•
Hemmecke and Schultz (2003)	•	•			•			•				•
Ahmed et al. (2004)		•	•		•			•		•	•	•
Sen and Higle (2005)				•		•	•		•			•
Kong et al. (2006)		•			•			•				•
Sen and Sherali (2006)				•		•	•		•			•
Sherali and Zhu (2006)		•		•		•	•		•	•		•
Sherali and Smith (2009)		•		•		•	•		•	•	•	•
Yuan and Sen (2009)		•		•		•	•		•		•	•
Ntaimo (2010)		•		•		•	•			•	•	
Özaltın et al. (2012)		•			•			•				•
Alonso-Ayuso et al. (2011)	•			•		•	•				•	•
Current Study		•			•			•				•

Table 1: Comparison of two-stage stochastic IP studies involving integer second-stage variables

4 Algorithmic Development

Two-stage stochastic IPs with integer variables in the second-stage are particularly difficult to solve because of the discontinuity and nonconvexity of the expected recourse function (Stougie 1987). For a fixed x, evaluating $Q(x, \omega)$ may involve the solution of an IP for every $\omega \in \Omega$. There has, however, been some success in solving problems similar to (P1), including those that consider random constraint matrices (Ahmed et al. 2004, Carøe and Tind 1998, Ntaimo 2010, Sen and Higle 2005, Sen and Sherali 2006, Sherali and Fraticelli 2002, Sherali and Smith 2009, Sherali and Zhu 2006, Yuan and Sen 2009), uncertain or nonlinear objectives (Ahmed et al. 2004, Alonso-Ayuso et al. 2003, 2011, Ntaimo and Sen 2008, Özaltın et al. 2012, Yuan and Sen 2009), and/or multiple stages (Alonso-Ayuso et al. 2003, 2011, Hemmecke and Schultz 2003). Table 1 summarizes the problem classes considered in each of these studies, including whether it allows for a multi-stage framework (MS), contains a computational study (CS), and information on first-stage variables, second-stage variables, and stochastic components. For an additional survey we refer to Guan (2005).

Of these, Kong et al. (2006) and Özaltın et al. (2012) are of particular note. Kong et al. (2006) solve (P1), also with nonnegative and integral data, using the reformulation of Ahmed et al. (2004), obtaining z for a large set of integral right-hand sides. The explicit storage of z for both stages limits their approach, resulting in the moderate number of rows (no more than seven) their method can handle. They solve instances with large extensive forms (e.g., around 2 million rows, 140 million variables and 300,000 scenarios); we are unaware of other approaches that can solve larger instances in terms of extensive forms of this class. Özaltın et al. (2012) generalizes the approach by Kong et al. (2006) to quadratic integer programs.

Like Schultz et al. (1998) and Hemmecke and Schultz (2003), we also show the existence of an optimal solution within a tractable finite set. However, our set is based upon the truncated integral monoid formed from select constraint matrix columns, and we utilize the partial value functions corresponding to these monoid elements to identify a global optimum to (P1). Like Ahmed et al. (2004), we also construct a global branch-and-bound approach that exploits the (partial) value function, though our approach optimizes over the truncated integral monoid. We also include means to eliminate suboptimal monoid elements prior to burdensome expectation computations.

The approach of Kong et al. (2006) is very efficient on certain test instances having extremely large extensive forms; however, as previously noted the test instances they can solve to optimality are unfortunately limited by memory to seven rows. We also make use of IP value function properties to obtain the value function in the following distinct ways. We remove unnecessary columns using both Corollary 1 and Proposition 6 prior to computing the truncated integral monoid. We also leverage integer complementary slackness algorithmically (Proposition 2) to extract certain monoid elements not satisfying Corollary 2; for such elements, the partial value function is then easily computed. Additionally, we take advantage of Proposition (c) (nondecreasing) in several key areas to enhance our algorithmic bounding strategies.

Building upon Theorem 7, we introduce here the components used in our search for a first-stage monoid element $\pi^* \in \bar{M}^1$ that solves (P3) and thus (P2). We generate and store the truncated integral monoids $\bar{M}^1$ and $\bar{M}^2$ along with their corresponding partial value functions $\bar{\psi}$ and $\bar{\phi}$ for both stages. We also present global branch-and-bound using one of three different bounding strategies that search for π^* over $\mathbf{B}^1$ and compare it with exhaustive search over a subset of $\bar{M}^1$.

The nonnegativity and integrality restrictions of Assumption **A6** ensure the finiteness of $\mathbf{B}^1$ and $\mathbf{B}^2$ (as our method requires storage of the partial value function in memory), and imply that $\beta_k \in \mathbb{Z}_+^m \ \forall \ \beta_k \in \mathbf{B}^k, k = 1, 2$. Also, $\bar{\mathbf{B}}^k \neq \emptyset$ because $0 \in \bar{\mathbf{B}}^k$, $k = 1, 2$. Thus, Assumptions **A1** and **A2** are satisfied for the value functions of both stages.

Let $\mathbf{u}_i = \min\limits_{\omega \in \Omega} h_i(\omega), \ \forall \ i = 1, \ldots, m$, and in light of the nonnegativity of T and W, define $\mathbf{B}^1 = \prod_{i=1}^m [0_i, \mathbf{u}_i] \cap \mathbb{Z}^m$. Then $\mathbf{B}^1$ is the set of vectors $\beta_1 \in \mathbb{Z}_+^m$ such that there exists a feasible first-stage $x \in S_1(\beta_1)$ with $\beta_1 \leq h(\omega)$ for all $\omega \in \Omega$. Let $\bar{h}_i = \max\limits_{\omega \in \Omega} h_i(\omega), \ \forall \ i = 1, \ldots, m$, then $\mathbf{B}^2 = \prod_{i=1}^m [0, \bar{h}] \cap \mathbb{Z}^m$. This finite construction of $\mathbf{B}^1$ and $\mathbf{B}^2$ yields $|\mathbf{B}^1| = \prod_{i=1}^m (\min\limits_{\omega \in \Omega} h_i(\omega) + 1)$ and $|\mathbf{B}^2| = \prod_{i=1}^m (\max\limits_{\omega \in \Omega} h_i(\omega) + 1)$.

4.1 Computing Truncated Integral Monoids and Partial Value Functions

For ease of exposition we use first-stage notation in this section to represent operations on the problems of both stages (i.e., first-stage objective coefficient vector c is used to represent both c and d, etc.), applying in identical fashion the corresponding operations to the second-stage problem.

Procedure 1 generates $\bar{M}^1$ and $\bar{\psi}$. Let set $\mathcal{L}^1$ contain $\pi \in \bar{M}^1 \subseteq \mathbf{B}^1$; for every $\pi \in \bar{M}^1$, we also simultaneously compute and maintain $\bar{\psi}(\pi)$. We leverage integer complementary slackness (Proposition 2) to yield $\bar{\psi}$ for additional $\pi \in \bar{M}^1$ without solving the corresponding IPs. We note that the two types of columns elimination discussed in Example 2.2 are applied to T prior to generating $\bar{M}^1$, yielding the $m \times \bar{n}_1$ matrix $\overline{T}$ indexed by $\overline{\mathcal{T}}$. We also derive upper bounds u_j on the variables. Section 7.1 of the e-companion contains more details on these associated operations.

Algorithm 1 Generate Integral Monoid and Partial Value Functions

1: **GENERATE MONOID** (j, λ)
2: **for** $k = u_j$ to 0 **do**
3: Construct the m-dimensional vector $\pi = \lambda + k \cdot t_j$.
4: **if** $\pi \in \mathbf{B}^1$ **then** {i.e., $\pi \leq \mathbf{u}$ }
5: **if** $j < \bar{n}_1$ **then**
6: Call **GENERATE MONOID** $(j+1, \pi)$.
7: **else**
8: **if** $\pi \notin \mathcal{L}^1$ **then**
9: Solve IP associated with $\bar{\psi}(\pi)$ to obtain optimal solution $\hat{x}$.
10: **if** $\overline{T}\hat{x} \neq \pi$ **then** {Corollary 2}
11: Do not store $\pi \in \mathcal{L}^1$.
12: **else**
13: **for all** $x \in \mathbb{Z}_+$ such that $x \leq \hat{x}$ (includes $x = \hat{x}$) **do**
14: Compute monoid element $\overline{T}x$.
15: **if** $\overline{T}x \notin \mathcal{L}^1$ **then**
16: Store $\overline{T}x$ and $\bar{\psi}(\overline{T}x) = c^\top x$ in $\mathcal{L}^1$. {ICS (Proposition 2)}

In Procedure 1, we recursively generate monoid elements using integer linear combinations of the columns t of $\overline{T}$. After initialization with $j = 1$ and $\lambda = 0$, Procedure 1 works in a downward fashion beginning with the variable upper bounds u_j (**Lines 1 − 6**). When we arrive at a monoid element $\pi \in \mathbf{B}^1$ that is not already contained in $\mathcal{L}^1$, we solve $z(\pi)$ (e.g., using CPLEX (ILOG 2007)) and obtain an optimal solution $\hat{x}$ (**Lines 8 − 9**). If $\overline{T}\hat{x} \neq \pi$, we discard π as per Corollary 2 (**Lines 10 − 11**). Otherwise, without solving additional IPs (**Lines 12 − 16**), we leverage Proposition 2 to discover additional $\pi \in \bar{M}^1$ for which $\bar{\psi}(\pi)$ is easily obtainable. That is, for all $x \leq \hat{x}$ we compute monoid element $\overline{T}x$; if $\overline{T}x \notin \mathcal{L}^1$, store $\overline{T}x$ and $\bar{\psi}(\overline{T}x) = c^\top x$ in $\mathcal{L}^1$. Otherwise $\overline{T}x \in \mathcal{L}^1$; we stop applying Proposition 2 due to the likelihood that the underlying structure has already been explored.

In the exposition that follows, we assume the values of $\psi(\mathbf{u})$ and $\phi(\bar{h})$ are known from solving the corresponding IPs. For every $\pi \in \bar{M}^1$ we also store either **(i)** the expectation over the second-stage value functions $\mathbb{E}\phi(h(\omega) - \pi)$ (if known) or **(ii)** an upper bound U_π on this expectation. We only maintain such information for $\mathcal{L}^1$.

Each generated $\pi \in \mathcal{L}^1$ is assigned a unique numerical identifier. As look-up efficiency is important to the overall run-time, we utilize binary search to quickly locate $\pi \in \mathcal{L}^1$ and associated $\bar{\psi}(\pi)$ by maintaining that $\mathcal{L}^1$ is sorted by this unique identifier.

4.2 Searching for an Optimal Monoid Element

In view of Theorem 7 we use a global branch-and-bound approach to search over $\mathbf{B}^1$ for an optimal monoid element π^* to (P3). We outline our approach in Procedure 2 (**GBB**).

At iteration k, we consider a single hyper-rectangle $\mathcal{H}^k = [l^k, u^k]$, focusing exclusively on $\pi \in \{\mathcal{L}^1 \cap \mathcal{H}^k\}$. To improve the efficiency of the algorithm, we introduce several enhancements in addition to the standard components of branching (**Lines 12 − 14**), fathoming (**Lines 8 – 9**), and identifying new incumbent solutions (**Lines 10 − 11**). We establish three distinct bounding approaches (**Lines 4, 16**; further detailed in Section 4.3 as well as Section 8 of the Appendices) that require only $\bar{\psi}$ and $\bar{\phi}$ to bound the optimal objective over $\mathcal{H}^k$. We also discuss the elimination of entire regions of $\mathbf{B}^1$ that contain no more than one $\pi \in \mathcal{L}^1$ in Section 9.2 of the e-companion.

Algorithm 2 Global Branch-and-Bound (**GBB**)

1: Construct the hyper-rectangle $\mathcal{H}^0 := [l^0, u^0] = \prod_{i=1}^m [0_i, \mathbf{u}_i]$.
2: Set $\mathcal{M} \leftarrow \{\mathcal{H}^0\}$, $\pi^* = 0$, and $L = \bar{\psi}(0) + \mathbb{E}\phi(h(\omega) - 0)$. {Section 8.1 of e-companion}
3: $\forall\ \pi \in \mathcal{L}^1$ set expectation bound $U_\pi = \mathbb{E}\phi(h(\omega) - 0)$. {Section 8.2 of e-companion}
4: Obtain initial bounds μ^0 and ν^0. {BOUND: Pass $\mathcal{H}^0$ and L to Procedure 3}
5: Set $k \leftarrow 1$.
6: **while** $\mathcal{M} \neq \emptyset$ **do**
7:     Select and delete from $\mathcal{M}$ a hyper-rectangle $\mathcal{H}^k := [l^k, u^k] = \prod_{i=1}^m [l_i^k, u_i^k]$.
8:     **if** $\nu^k \leq L$ **then** {FATHOM}
9:         Fathom $\mathcal{H}^k$, that is, $\mathcal{M} \leftarrow \mathcal{M} \setminus \{\mathcal{H}^k\}$.
10:     **else if** $l^k = u^k$ and $L < \mu^k$ **then** {NEW INCUMBENT}
11:         Update $L = \mu^k = \nu^k = f^k$, and set $\pi^* = l^k = u^k$.
12:     **else** {BRANCH}
13:         Choose an index i', $1 \leq i' \leq m$, such that $l_{i'}^k < u_{i'}^k$.
14:         Divide $\mathcal{H}^k$ into $\mathcal{H}^{k_1}$ and $\mathcal{H}^{k_2}$ along index i' by a user-defined disjunction.
15:         **for** $\ell = 1, 2$ **do**
16:             Obtain bounds μ^{k_ℓ} and ν^{k_ℓ}. {BOUND: Pass $\mathcal{H}^{k_\ell}$ and L to Procedure 3}
17:             Add $\mathcal{H}^{k_\ell}$ to $\mathcal{M}$, i.e., $\mathcal{M} \leftarrow \mathcal{M} \cup \{\mathcal{H}^{k_\ell}\}$.
18:     Remove suboptimal $\pi \in \mathcal{L}^1$ using U_π bound. {Section 9.1 of e-companion}
19:     Set $k \leftarrow k + 1$.
20: Terminate with optimal solution π^*.

Because one of the main bottlenecks in solving two-stage stochastic programs is computing the expectation over all scenarios, we discuss means to potentially avoid this computational burden. For any $\pi \in \mathcal{L}^1$, we introduce U_π so that

$$U_\pi \geq \mathbb{E}\phi(h(\omega) - \pi), \tag{10}$$

holding at equality when the exact value $\mathbb{E}\phi(h(\omega) - \pi)$ is known. We first compute the exact value of $\mathbb{E}\phi(h(\omega))$ corresponding to $0 \in \mathcal{L}^1$ (**Line 2**) and use this expectation to initialize U_π for all $\pi \in \mathcal{L}^1$ by the nondecreasing property of the value function (**Line 3**; further detailed in Section 8.2 of the e-companion). During calls to Procedure 3 (**Lines 4, 16**) the U_π values are subsequently tightened (**Line 9** of Procedure 3). The main purpose of the U_π values is to avoid the explicit calculation of the expectation through a bounding argument, providing for the elimination of suboptimal $\pi \in \mathcal{L}^1$ (**Line 18**; further details in Sections 8.2 and 9.1 of the e-companion).

Any existing truncated integral monoid element $\pi \in \mathcal{L}^1$ has not been shown to be suboptimal, so that at all times, $\mathcal{L}^i \subseteq \bar{M}^i$, $i = 1, 2$. Also, let $\mathcal{M}$ be a set of unfathomed hyper-rectangles; each $\mathcal{H}^k \in \mathcal{M}$ is associated with a subproblem of the form

$$f^k = \max_{\pi \in \left\{\mathcal{L}^1 \cap \mathcal{H}^k\right\}} \left\{\bar{\psi}(\pi) + \mathbb{E}\phi(h(\omega) - \pi)\right\}, \tag{11}$$

a lower bound $\mu^k \leq f^k$, and an upper bound $\nu^k \geq f^k$. A global lower bound L represents the incumbent objective value derived from some $\pi \in \mathcal{L}^1$ for which $\bar{\psi}(\pi) + \mathbb{E}\phi(h(\omega) - \pi)$ is exactly known.

4.3 Calculating Bounds on Hyper-Rectangle $\mathcal{H}^k$

Procedure 3 outlines our approach to bound f^k for $\mathcal{H}^k$. We construct bounds μ^k and ν^k on f^k using $\pi \in \mathcal{L}^1$ and $\tau \in \mathcal{L}^2$ with corresponding $\bar{\psi}(\pi)$ and $\bar{\phi}(\tau)$. Our approach differs from that of Kong et al. (2006), which relies upon l^k and u^k, the corners of hyper-rectangle $\mathcal{H}^k$, to generate simple bounds $\bar{\mu}^k$ and $\bar{\nu}^k$ on f^k:

$$\bar{\mu}^k = \psi(l^k) + \mathbb{E}\phi(h(\omega) - u^k), \quad \text{and} \quad \bar{\nu}^k = \psi(u^k) + \mathbb{E}\phi(h(\omega) - l^k). \tag{12}$$

Because typically $l^k \notin \bar{M}^1$ and $u^k \notin \bar{M}^1$, these bounds require storing in memory the value function for the entire domain (i.e., all $\beta \in \mathbf{B}^i,\ i = 1, 2$). As the number of rows increases, $|\mathbf{B}^i|,\ i = 1, 2$ increase exponentially in size, thereby substantially reducing the effectiveness of their algorithm. We instead exploit $\bar{\psi}$ and $\bar{\phi}$ to bound f^k, which typically requires far less memory; also, if $\{\mathcal{L}^1 \cap \mathcal{H}^k\} = \emptyset$, we can directly fathom $\mathcal{H}^k$ (**Lines 3 − 4**). Specifically, we use the lower bound

$$\mu^k = \max_{\pi \in \mathcal{R}_1^k} \left\{\bar{\psi}(\pi) + \mathbb{E}\phi(h(\omega) - \pi)\right\}, \tag{13}$$

where $\mathcal{R}_1^k \subseteq \{\mathcal{L}^1 \cap \mathcal{H}^k\}$ is defined as the set of $\pi \in \mathcal{L}^1$ contained within $\mathcal{H}^k$ having *known* $\mathbb{E}\phi(h(\omega) - \pi)$ values (we ensure that $\mathcal{R}_1^k \neq \emptyset$ in **Lines 7 − 8** of Procedure 3). We next briefly describe three upper bounds that we implement in our algorithm which are visually depicted in Figure 1.

The first upper bound is given by:

$$\nu_{I1}^k = \max_{\pi \in \left\{\mathcal{L}^1 \cap \mathcal{H}^k\right\}} \left\{\bar{\psi}(\pi) + U_\pi\right\}, \tag{14}$$

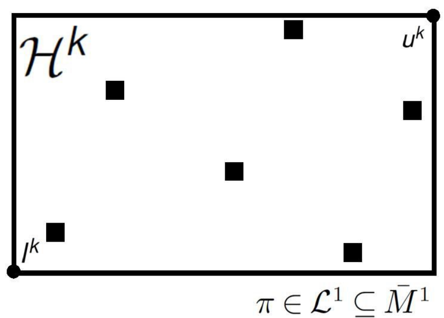
where U_π is defined in (10). Next consider $\pi \in \mathcal{L}^1$ that is nearest to l^k with respect to dimension i', yet smaller, i.e. $\pi \in \mathcal{L}^1$ such that $\pi_i \leq l_i^k$ for all dimensions i. While any choice of i' could work, we choose a dimension $i' \in_{i \in \{1,\ldots,m\}} l_i^k$. Let $\mathcal{N}_l^k$ be the set containing all such $\pi \in \mathcal{L}^1$. Then the second upper bound is given by:

$$\nu_{E1}^k = \max_{\pi \in \left\{\mathcal{L}^1 \cap \mathcal{H}^k\right\}} \bar{\psi}(\pi) + \min\left\{\min_{\pi \in \mathcal{N}_l^k} U_\pi,\ \phi(\bar{h})\right\}. \tag{15}$$

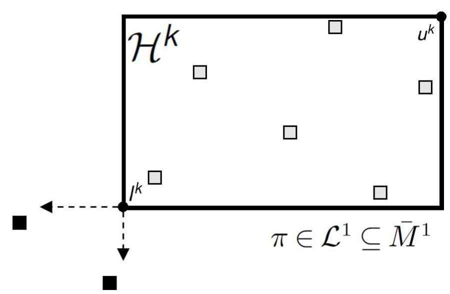
For each scenario $\omega \in \Omega$, let $\delta^\omega = h(\omega) - l^k$ (consistent with Assumption **A5**, by the construction of $\mathbf{u}$ we assume $h(\omega) \geq l^k \; \forall \; \omega \in \Omega$, ensuring $\delta^\omega \geq 0$). Consider $\tau^\omega \in \mathcal{L}^2$ that is nearest to δ^ω with respect to dimension i', yet larger, i.e. $\tau^\omega \in \mathcal{L}^2$ such that $\tau_i^\omega \geq \delta_i^\omega$ for all dimensions i. Over all dimensions $i = 1, \ldots, m$, we choose dimension $i' \in_{i \in \{1,\ldots,m\}} \delta_i^\omega$ (again, any choice of i' could work). Let $\mathcal{N}_{\delta^\omega}^k$ be the set containing all such $\tau^\omega \in \mathcal{L}^2$. This definition provides the third upper bound:

$$\nu_{E2}^k = \max_{\pi \in \left\{\mathcal{L}^1 \cap \mathcal{H}^k\right\}} \bar{\psi}(\pi) + \mathbb{E} \min \left\{ \min_{\tau^\omega \in \mathcal{N}_{\delta^\omega}^k} \phi(\tau^\omega), \; \phi(\bar{h}) \right\}. \tag{16}$$

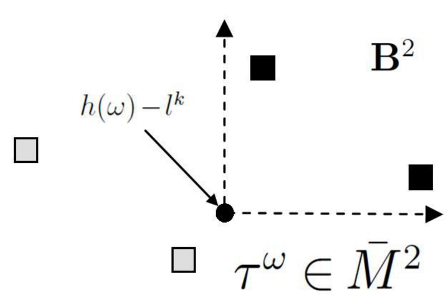
A more in-depth discussion of these bounds and their validity can be found in Section 8 of the e-companion (lower bounding addressed in Section 8.3; upper bounding in Section 8.4).



(a) ν_{I1}^k



(b) ν_{E1}^k



(c) ν_{E2}^k

Figure 1: Symbol ■ depicts $\pi \in \mathcal{L}^1 \cap \mathcal{H}^k$ in (a), $\pi \in \mathcal{L}^1$ s.t. $\pi \leq l^k$ in (b), and $\tau^\omega \in \bar{M}^2$ s.t. $\tau^\omega \geq h(\omega) - l^k$ in (c).

Theorem 8 ***GBB*** *terminates with an optimal solution to* (P2) *after a finite number of steps.*

Theorem 8 follows directly from finite consistency arguments (see, e.g., Horst and Tuy (1996)).

Algorithm 3 Bound $\mathcal{H}^k$

Input: Hyper-rectangle $\mathcal{H}^k$, global lower bound L.
Initialize $\mu^k = -\infty$ and $\nu^k = +\infty$.
if $\left\{\mathcal{L}^1 \cap \mathcal{H}^k\right\} = \emptyset$ **then** {FATHOM}
  Fathom $\mathcal{H}^k$.
else
  Partition $\left\{\mathcal{L}^1 \cap \mathcal{H}^k\right\}$ into $\mathcal{R}_1^k$, $\mathcal{R}_2^k$ (π with known and unknown expectations, respectively).
  if $|\mathcal{R}_1^k| < |\mathcal{L}^1 \cap \mathcal{H}^k|$ **then** {FIND ONE EXACT EXPECTATION}
    Choose $\hat{\pi} \in \mathcal{R}_2^k$ and compute $\mathbb{E}\phi(h(\omega) - \hat{\pi})$. {Section 8.1 of e-companion}
    $\forall\, \pi \in \mathcal{L}^1 : \pi \geq \hat{\pi}$, update $U_\pi = \min\left\{U_\pi, \mathbb{E}\phi(h(\omega) - \hat{\pi})\right\}$. {Section 8.2 of e-companion}
    if $\bar{\psi}(\hat{\pi}) + \mathbb{E}\phi(h(\omega) - \hat{\pi}) > L$ **then**
      $L = \bar{\psi}(\hat{\pi}) + \mathbb{E}\phi(h(\omega) - \hat{\pi})$. {NEW INCUMBENT: $\hat{\pi}$ IMPROVES L}
    Set $\mathcal{R}_2^k \leftarrow \mathcal{R}_2^k \backslash \hat{\pi}$, and set $\mathcal{R}_1^k \leftarrow \mathcal{R}_1^k \cup \hat{\pi}$.
  if $|\mathcal{L}^1 \cap \mathcal{H}^k| = 1$ **then** {SEARCH SPACE REDUCTION}
    Set $l^k = u^k = \pi \in \left\{\mathcal{L}^1 \cap \mathcal{H}^k\right\}$ {i.e., $|\mathcal{L}^1 \cap \mathcal{H}^k| = 1$, so reduce $\mathcal{H}^k$ to single monoid element}.
    Set $\mu^k = \nu^k = \bar{\psi}(\pi) + \mathbb{E}\phi(h(\omega) - \pi)$.
  else
    $\mu^k = \max_{\pi \in \mathcal{R}_1^k} \left\{\bar{\psi}(\pi) + \mathbb{E}\phi(h(\omega) - \pi)\right\}$.
    Determine ν^k using one of three bounds. {SECTION 8.4 of e-companion}
Output: Bounds μ^k and ν^k.

4.4 Exhaustive Search over $\mathcal{L}^1$

We also consider exhaustive search over a subset of $\bar{M}^1$ as an alternative to global branch-and-bound. This approach uses $\mathcal{L}^1$ and $\mathcal{L}^2$ as well as $\bar{\psi}$ and $\bar{\phi}$ generated in Section 4.1. For each $\pi \in \mathcal{L}^1$, the expectation is evaluated according to Section 8.1 of the e-companion, terminating after iterating over all $\pi \in \mathcal{L}^1$ to identify the optimal solution π^*.

5 Computational Experiments

We compare four algorithmic approaches in the search for a global optimum π^* to (P3). The first three involve global branch-and-bound using the three upper bound strategies described in Sections 4.3 and 8.4 of the Appendices, while the fourth is the Exhaustive Search over

$\mathcal{L}^1$ of Section 4.4. The algorithms are implemented in C++, compiled using Microsoft Visual Studio 2005 and tested on a PC equipped with Windows XP, a dual-core Intel 3GHz processor, 3GB of RAM, and CPLEX 11.0 (ILOG 2007).

5.1 Test Strategies and Instance Classes

We propose two testing strategies to gain computational insights. The first strategy explores the performance of our algorithms on test instance classes labeled IC-T**X** that vary several key parameters, including the number of rows, scenarios, columns, and constraint matrix densities. The second testing strategy benchmarks our approaches against test instance classes labeled IC-K**X** that are patterned after those in Kong et al. (2006).

We detail the sizes of our considered test instance classes in Table 2, which includes the number of extensive-form constraints and variables, as well as the sizes of $|\mathbf{B}^1|$ and $|\mathbf{B}^2|$. We design all of our test instances to satisfy Assumptions **A3** – **A6**; all parameters are chosen to parallel levels in Kong et al. (2006). We use a Bernoulli distribution to model density Δ (of T and W), varying its values over $\{0.3, 0.4, 0.5\}$. Coefficients d_j and w_{ij} are uniformly distributed in the intervals $[2, 6]$ and $[2, 5]$, respectively. Likewise, coefficients c_j and t_{ij} are uniformly distributed in the intervals given by "min|max" in Table 2. The m entries of the $h(\omega)$ vector are independently generated and uniformly distributed according to the intervals $[5, 9]$ (for IC-K1 through IC-K6) and $[5, 10]$ (for IC-K7 through IC-K10 and all IC-T**X**). All scenarios are verified to be unique. Our test instances are available online (Trapp 2012) in SMPS format (Gassmann 1990, 2005). For every instance class we consider, three test instances are generated for the first strategy and one for the second strategy.

We also evaluate the effect of a more granularly distributed ω on computational run-time. We have designed instances IC-T13, IC-T14, and IC-T15 to reflect a ten-fold increase in $|\Omega|$ over related instances IC-T3, IC-T5, and IC-T7 (holding all other parameters constant), in order to evaluate how our algorithmic performance scales with $|\Omega|$. The particular levels of $|\Omega|$ we choose in Table 2 are comparable to those appearing in related studies, although we note that the scenarios for IC-K7 through IC-K10 have been adjusted to $|\Omega| = 7,812$ from $|\Omega| = 279,936$ in Kong et al. (2006). In the discussions that follow we mostly refer to their study due to comparable extensive-form sizes.

As we highlighted in Section 4, various techniques have been developed to solve stochastic integer programs. We are unaware, however, of any approaches that solve larger instances of the considered class with respect to the simultaneous combination of rows m, variables n_1 and n_2, and scenarios $|\Omega|$ as represented in the extensive-form sizes in Table 2. For example, using the approach of Kong et al. (2006) to solve some of the instances we propose (e.g., IC-T16 with $m = 250$ rows) would be prohibitively expensive, requiring approximately 2E260 MB of memory. Thus our approach may best be viewed as complementary in nature when compared with others in the literature.

5.2 Computational Results

We next provide an analysis of our computational findings for the test instances in Table 2.

	$\|\Omega\|$	m	n_1	n_2	Δ	min|max c_j	min|max t_{ij}	Extensive Form Rows	Extensive Form Cols	Search Space Size $\|\mathbf{B}^1\|$	Search Space Size $\|\mathbf{B}^2\|$
IC-K1	7,812	6	1000	100	0.5	1|5	1|4	4.7E4	7.8E5	4.7E4	1.0E6
IC-K2	7,812	6	500	500	0.5	1|5	1|4	4.7E4	3.9E6	4.7E4	1.0E6
IC-K3	7,812	6	300	500	0.5	1|5	1|4	4.7E4	3.9E6	4.7E4	1.0E6
IC-K4	7,812	6	500	500	0.5	1|10	1|4	4.7E4	3.9E6	4.7E4	1.0E6
IC-K5	7,812	6	500	500	0.5	1|5	1|3	4.7E4	3.9E6	4.7E4	1.0E6
IC-K6	7,812	6	500	500	0.8	1|5	1|4	4.7E4	3.9E6	4.7E4	1.0E6
IC-K7	7,812	7	1000	500	0.7	1|5	1|4	5.5E4	3.9E6	2.8E5	1.9E7
IC-K8	7,812	7	500	500	0.7	1|5	1|4	5.5E4	3.9E6	2.8E5	1.9E7
IC-K9	7,812	7	300	500	0.7	1|5	1|4	5.5E4	3.9E6	2.8E5	1.9E7
IC-K10	7,812	7	1000	500	0.3	1|5	1|4	5.5E4	3.9E6	2.8E5	1.9E7
IC-T1	1,000	20	50	50	0.3	1|5	1|4	2.0E4	5.0E4	3.7E15	6.7E20
IC-T2	1,000	20	50	50	0.4	1|5	1|4	2.0E4	5.0E4	3.7E15	6.7E20
IC-T3	1,000	20	50	50	0.5	1|5	1|4	2.0E4	5.0E4	3.7E15	6.7E20
IC-T4	5,000	50	100	100	0.3	1|5	1|4	2.5E5	5.0E5	8.1E38	1.2E52
IC-T5	5,000	50	100	100	0.4	1|5	1|4	2.5E5	5.0E5	8.1E38	1.2E52
IC-T6	5,000	50	100	100	0.5	1|5	1|4	2.5E5	5.0E5	8.1E38	1.2E52
IC-T7	8,000	100	200	200	0.3	1|5	1|4	8.0E5	1.6E6	6.5E77	1.4E104
IC-T8	8,000	100	200	200	0.4	1|5	1|4	8.0E5	1.6E6	6.5E77	1.4E104
IC-T9	8,000	100	200	200	0.5	1|5	1|4	8.0E5	1.6E6	6.5E77	1.4E104
IC-T10	10,000	100	300	300	0.3	1|5	1|4	1.0E6	3.0E6	6.5E77	1.4E104
IC-T11	10,000	100	300	300	0.4	1|5	1|4	1.0E6	3.0E6	6.5E77	1.4E104
IC-T12	10,000	100	300	300	0.5	1|5	1|4	1.0E6	3.0E6	6.5E77	1.4E104
IC-T13	10,000	20	50	50	0.5	1|5	1|4	2.0E5	5.0E5	3.7E15	6.7E20
IC-T14	50,000	50	100	100	0.4	1|5	1|4	2.5E6	5.0E6	8.1E38	1.2E52
IC-T15	80,000	100	200	200	0.3	1|5	1|4	8.0E6	1.6E7	6.5E77	1.4E104
IC-T16	300,000	250	1000	500	0.3	1|5	1|4	7.5E7	1.5E8	3.4E194	2.2E260

Table 2: Dimensions of test instance classes considered; $|\mathbf{B}^1|$, $|\mathbf{B}^2|$ are the number of right-hand sides in $\mathbf{B}^1$, $\mathbf{B}^2$

	$\lvert\mathcal{L}^1\rvert$	$\lvert\mathcal{L}^2\rvert$	IPs	Time (in seconds) CE	PVFC	RDM	Total
IC-T1	18,860	37,680	30,515	< 1	47	119	166
IC-T2	1,037	1,407	1,315	< 1	1	9	11
IC-T3	227	412	408	< 1	< 1	< 1	1
IC-T4	2,525	3,846	3,354	2	5	2	8
IC-T5	338	450	554	3	1	< 1	4
IC-T6	139	190	253	4	< 1	< 1	4
IC-T7	886	986	1,326	18	4	< 1	23
IC-T8	250	270	415	30	2	< 1	32
IC-T9	201	201	398	46	2	< 1	48
IC-T10	2,094	4,020	4,719	40	20	3	63
IC-T11	389	457	663	68	4	< 1	72
IC-T12	302	306	595	106	4	< 1	110
IC-T13	280	360	394	< 1	< 1	< 1	1
IC-T14	279	496	521	3	1	< 1	4
IC-T15	1,057	1,039	1,510	19	5	< 1	24
IC-T16	1,094	537	1,439	1,623	43	< 1	1,665

Table 3: Performance summary for partial value function computation and associated operations

5.2.1 Analysis of Partial Value Function Computation Performance

Table 3 reports computational results for computing $\bar{\psi}$ and $\bar{\phi}$ and associated operations (see Section 4.1, also 7 of the e-companion) for the 16 IC-T**X** test instance classes of Table 2. It includes the initial sizes of $|\mathcal{L}^i|$, $i = 1, 2$, the number of CPLEX 11 (ILOG 2007) calls ("IPs"), as well as time spent in column elimination ("CE"), partial value function calculations ("PVFC"), removing dominating elements from level sets ("RDM"), and total time ("Total"). All running times are averaged over three test instances for each instance class, and are cumulative over both stages.

The time spent generating $\mathcal{L}^1$ and $\mathcal{L}^2$ appears comparable to the time spent for the more efficient DP-based approach used to generate the value function in Kong et al. (2006). For an instance that is at least as large (and with a considerably greater number of rows), the longest cumulative time in their study was 1,226 seconds compared to 1,665 seconds in our study.

5.2.2 Analysis of Search Algorithm Performance

Table 4 presents the performance of our four solution approaches in finding a global optimum for the 16 IC-T**X** test instance classes of Table 2. We report the time spent in global branch-and-bound for three distinct bounding strategies (further referred to as I1, E1, and E2, see Section 4.3 and Section 8.4 of the Appendices), as well as the total execution time of the algorithm; only the latter is reported for Exhaustive Search. Averaged run-times are

	Bound I1		Bound E1		Bound E2		Exhaustive
	GBB	Total	GBB	Total	GBB	Total	Total
IC-T1	27:04:21	27:07:40	27:00:21	**27:03:39**	43:58:52	44:02:12	†49:27:36
IC-T2	0:14:52	0:14:56	0:14:39	**0:14:42**	0:21:32	0:21:36	0:49:03
IC-T3	0:01:28	0:01:29	0:01:27	**0:01:28**	0:02:27	0:02:28	0:03:09
IC-T4	9:19:31	**9:19:49**	9:20:57	9:21:15	11:51:42	11:52:00	29:38:12
IC-T5	0:16:17	0:16:22	0:16:01	**0:16:07**	0:24:27	0:24:32	0:28:28
IC-T6	0:04:11	0:04:16	0:04:08	**0:04:13**	0:07:01	0:07:07	0:05:05
IC-T7	2:12:03	**2:12:34**	2:12:18	2:12:49	2:46:46	2:47:17	4:35:45
IC-T8	0:20:51	0:21:27	0:20:51	**0:21:26**	0:30:17	0:30:52	0:23:22
IC-T9	0:13:23	0:14:15	0:13:23	0:14:15	0:19:54	0:20:46	**0:14:11**
IC-T10	22:09:03	22:10:30	22:00:28	**22:01:55**	27:04:46	27:06:13	53:02:41
IC-T11	1:03:23	1:04:41	1:03:06	**1:04:24**	1:28:21	1:29:38	1:16:26
IC-T12	0:37:46	0:39:42	0:37:22	0:39:18	0:54:01	0:55:57	**0:39:17**
IC-T13	0:17:27	0:17:28	0:17:12	**0:17:14**	0:30:23	0:30:24	0:33:10
IC-T14	2:32:52	2:33:00	2:32:22	**2:32:29**	3:45:23	3:45:30	4:10:03
IC-T15	27:17:18	27:17:58	27:09:55	**27:10:35**	35:10:11	35:10:51	58:08:53
IC-T16	150:44:03	**151:34:03**	150:48:34	151:38:36	186:42:01	187:31:57	163:00:51

Table 4: Comparison of performance for four algorithms on IC-T**X** instances

formatted as hh:mm:ss. The fastest total times for each test instance class are bolded. A computational run-time threshold of 200 hours was implemented; †indicates that two of three test instances exceeded this threshold.

Across all of our algorithms we observe that the vast majority of the overall running time is spent searching for an optimal solution π^* to (P3), with only a small fraction (around 0.5%) spent in partial value function computation and associated operations. Of particular note is instance IC-T1, which experienced rather lengthy run-times with upper bounds I1, E1 and E2, and did not finish within the allowable threshold for two instances using Exhaustive Search. Though its instances belong to our smallest considered class (in terms of initial problem dimensions), it had by far the most first-stage and second-stage monoid elements to consider. This sheds light on a broader and intuitive trend revealed through our computational studies: holding all other parameters equal, overall run-times are roughly proportional to the initial sizes of $|\mathcal{L}^1|$ and $|\mathcal{L}^2|$.

In contrasting the four algorithms, E2 is the only variant that was uniformly dominated by the others. However, among the other three, the results indicate that both I1 and E1 have nearly identical performance and tend to outperform Exhaustive Search. We note that when Exhaustive Search does slightly outperform the others, it is when there are few monoid elements in $\mathcal{L}^1$ and $\mathcal{L}^2$. We attribute this to the extra overhead required to set up the global branch-and-bound framework for I1 and E1.

	$\|\mathcal{L}^1\|$	$\|\mathcal{L}^2\|$	IPs	CE	PVFC	RDM	GBB	Total
IC-K1	46,656	225	49	0:00:02	1:44:52	0:00:46	2:00:03	3:46:04
IC-K2	–	–	–	–	–	–	–	–
IC-K3	6,480	6,480	81,882	0:00:01	0:01:21	0:00:02	11:55:45	11:57:12
IC-K4	15,552	12,920	114,231	0:00:01	0:01:47	0:00:07	6:42:38	6:44:41
IC-K5	31,104	8,064	36,144	0:00:01	0:03:39	0:00:31	52:17:51	52:22:17
IC-K6	2,258	2,993	36,407	0:00:02	0:01:33	0:00:01	3:51:36	3:53:13
IC-K7	24,378	7,685	181,016	0:00:04	0:05:38	0:00:30	48:18:27	48:24:51
IC-K8	10,549	15,399	377,785	0:00:02	0:25:22	0:00:21	57:25:03	57:50:54
IC-K9	5,781	3,644	48,623	0:00:01	0:01:38	0:00:04	12:17:23	12:19:10
IC-K10	–	–	–	–	–	–	–	–

Table 5: Performance of E1 on IC-K**X** instances: "–" indicates that the threshold of 200 hours was exceeded

5.2.3 Comparison With an Explicit Value Function Enumeration Approach

To understand how our best algorithm performs on test instance classes similar to others in the literature, we tested Variant E1 on the IC-K**X** test instance classes. We first note the sizes of $\mathcal{L}^1$ and $\mathcal{L}^2$ as compared to the IC-T**X** instances reported in Table 3. This is indicative of the general trend that instances with smaller values of m tend to have more monoid elements. In fact, it is quite likely that the threshold of 200 hours would have been reached before instances IC-K7 through IC-K10 had completed had we not reduced $|\Omega|$ from 279,936 to 7,812.

As detailed in Table 5, we were able to solve all but one of instances IC-K1 through IC-K6 in reasonable times. The exception was IC-K2, where T contains a matrix similar to identity submatrix I (see Proposition 9), causing $|\mathcal{L}^1| = \frac{1}{2}|\mathbf{B}^1|$. For test instances IC-K1 and IC-K10, T contains I (so that $|\mathcal{L}^1| = |\mathbf{B}^1|$), and in general is indicative of a potential computational limitation of our characterization. In view of the run-times reported in Kong et al. (2006), it appears that when the complete value function can be stored in memory, their approach outperforms ours. However, for similar reasons, their approach fails on all of our test instances IC-T**X**, so these approaches may be viewed as complementary.

5.2.4 Discussion

In observing the overall performance, we notice that run-times are roughly proportional to both the initial sizes of $|\mathcal{L}^1|$ and $|\mathcal{L}^2|$, as well as $|\Omega|$. Across every instance class and algorithmic approach, as the number of monoid elements increase, so do the run-times. We also observe the general tendency of the number of monoid elements to increase when constraint matrix densities decrease, numbers of rows m decrease, numbers of columns n_1 and n_2 increase, and the value $\max_{\omega,i} h_i(\omega)$ increases.

When comparing the performance of our algorithms on test instance classes IC-T13, IC-T14, and ICT-15 to IC-T3, IC-T5, and IC-T7, we observe an average increase of roughly 12, 9, and 12 for the overall run-times; this increase is consistent across all four algorithms.

Because the only distinction between these three pairs of test instance classes was increasing $|\Omega|$ by a factor of 10, this suggests a roughly linear increase in algorithmic run-times as $|\Omega|$ increases, which was also noted in the approach of Hemmecke and Schultz (2003).

We observe that the maximum memory usage in our largest instances was well below 300MB. We attribute this to the relatively modest size of $|\mathcal{L}^1|$ and $|\mathcal{L}^2|$ as compared to the enormity of $|\mathbf{B}^1|$ and $|\mathbf{B}^2|$. We also observe that the number of monoid elements eliminated before calculation of an exact expectation (see 9.1) is usually more than 80% and in some cases as much as 98%. This likely had a significant contribution to the observed performance of the global branch-and-bound approach.

6 Concluding Remarks

We present a characterization of the value function of a pure linear integer program with inequality constraints. We introduce the set of level-set minimal vectors $\bar{\mathbf{B}}$ and demonstrate some of its theoretical properties, showing that the value function can be described using only its values among elements of $\bar{\mathbf{B}}$. Membership in $\bar{\mathbf{B}}$ is difficult to verify, so we instead develop our characterization over the truncated integral monoid $\bar{M} \supseteq \bar{\mathbf{B}}$.

We subsequently show how our characterization can be exploited to solve two-stage stochastic pure integer programs via a value function reformulation. Specifically, we develop an algorithmic framework that takes advantage of the underlying integral monoid to solve two-stage multidimensional knapsack problems with random budgets. While computational limitations exist for our approach (in particular, see Propositions 8 and 9), our experiments demonstrate that we are able to solve instances of this class with extensive forms that are among the largest instances solved in the literature, e.g., see Kong et al. (2006) and Özaltın et al. (2012), and with considerably more rows. As previously noted, value function reformulations (9) and (P3) as well as Theorems 6 and 7 do not require the nonnegativity restriction. Thus, our algorithmic development can potentially be extended to accommodate negative data, as long as the finiteness of $\mathbf{B}^1$ and $\mathbf{B}^2$ is maintained (recall that our method requires storage of the partial value function in memory). We leave this extension as a topic for future research.

Furthermore, our approach may be amenable to solve more general two-stage stochastic integer programs as long as the scenarios are divided into relatively few groups sharing the same d, T, and W. We also note that $\bar{M}$ is just one approximation of $\bar{\mathbf{B}}$, and in general, it may be possible to identify tighter supersets. Other open questions include considering the mixed-integer case and extending the problem to multistage stochastic integer programs.

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Additional Algorithmic and Computational Details

7 Computing Partial Value Functions and Associated Operations

We use first-stage notation in this section for operations on the problems of both stages.

7.1 Preprocessing

Column Elimination. We preprocess T to remove unnecessary columns, motivated by Corollary 1 as well as Proposition 6. We are then left with columns t_j of T that both **(i)** satisfy $\psi(t_j) = c_j$ and **(ii)** are level-set minimal vectors, yielding the $m \times \bar{n}_1$ matrix $\overline{T}$ indexed by $\overline{\mathcal{T}}$.

Variable Upper Bounds. We derive upper bounds on the variables to obtain the truncated integral monoid $\bar{M}^1$ corresponding to the columns of $\overline{T}$. Recalling that $\mathbf{u}_i = \min\limits_{\omega \in \Omega} h_i(\omega),\ \forall\, i = 1, \ldots, m$, we compute the upper bound u_j on x_j given by the closed form $u_j = \min\limits_{i: t_{ij} \neq 0} \lfloor \mathbf{u}_i / t_{ij} \rfloor$.

7.2 Postprocessing: Removing Level-Set Dominating Monoid Elements

Within distinct objective function level sets, we further process $\mathcal{L}^1$ by removing any monoid elements that are dominating, as by Definition 1 they clearly are not level-set minimal.

8 Calculating Bounds on Hyper-Rectangle $\mathcal{H}^k$

For iteration k, our bounding step (Procedure 3) uses two intermediate sets to bound $\mathcal{H}^k$. Recall that $\mathcal{R}_1^k \subseteq \{\mathcal{L}^1 \cap \mathcal{H}^k\}$ is defined as those $\pi \in \mathcal{L}^1$ contained within $\mathcal{H}^k$ with *known* $\mathbb{E}\phi(h(\omega) - \pi)$ values (we ensure that $\mathcal{R}_1^k \neq \emptyset$ in **Lines 7 − 8** of Procedure 3). Also, let $\mathcal{R}_2^k = \{\mathcal{L}^1 \cap \mathcal{H}^k\} \setminus \mathcal{R}_1^k$. For any $\mathcal{H}^k$ we assume $\{\mathcal{L}^1 \cap \mathcal{H}^k\} \neq \emptyset$ (addressed in **Lines 3 − 4** of Procedure 3).

8.1 Calculate Expectation for $\pi \in \mathcal{L}^1$

All of our solution approaches require exact expectation calculation (see **Line 2** of **GBB**, **Line 8** of Procedure 3, and Section 4.4). For a given $\pi \in \mathcal{L}^1$, we compute $\mathbb{E}\phi(h(\omega) - \pi)$ using only $\bar{\phi}$. To see this, for a given $\pi \in \mathcal{L}^1$ and some $\omega \in \Omega$, let $\delta^\omega = h(\omega) - \pi$. If $\delta^\omega \in \bar{M}^2$, $\bar{\phi}(\delta^\omega)$ is contained in $\mathcal{L}^2$. If $\delta^\omega \in \mathbf{B}^2 \backslash \bar{M}^2$ then $\delta^\omega \notin \bar{\mathbf{B}}^2$, and so by Lemma 5 there exists a $\tau^\omega \in \bar{M}^2$ for which $\bar{\phi}(\tau^\omega) = \phi(\delta^\omega)$. Thus we can locate τ^ω to serve as a surrogate for δ^ω. To determine $\phi(\delta^\omega)$, because ϕ is nondecreasing, we search over all $\tau^\omega \in \mathcal{L}^2$ with $\tau^\omega \leq \delta^\omega$. Among all such τ^ω, we assign the largest value function to $\phi(\delta^\omega)$. For $\pi \in \mathcal{L}^1$ we compute the exact value for the expectation $\mathbb{E}\phi(h(\omega) - \pi)$ as $\sum\limits_{\omega \in \Omega} p_\omega \phi(\delta^\omega)$.

8.2 Propagate Expectation Bound Over $\mathcal{L}^1$

The exact computation of $\mathbb{E}\phi(h(\omega) - \pi)$ as described in Section 8.1 is burdensome. After initially calculating the exact expectation in **GBB** for $0 \in \mathcal{L}^1$, they are subsequently computed only on an as-needed basis; instead, whenever possible we use a direct bounding strategy based on the U_π values to potentially eliminate suboptimal $\pi \in \mathcal{L}^1$ (**Line 18** of **GBB**).

Given $\hat{\pi} \in \mathcal{L}^1$, then $\forall\, \pi \in \mathcal{L}^1$ such that $\pi \geq \hat{\pi}$, $\mathbb{E}\phi(h(\omega) - \hat{\pi}) \geq \mathbb{E}\phi(h(\omega) - \pi)$.

Proof. The partial value function $\phi(h(\omega) - \pi)$ is nondecreasing with respect to $(h(\omega) - \pi)$. Taking the expectation gives $\sum_{\omega \in \Omega} p_\omega \cdot \phi(h(\omega) - \hat{\pi}) \geq \sum_{\omega \in \Omega} p_\omega \cdot \phi(h(\omega) - \pi)$. ■

Given any $\hat{\pi} \in \mathcal{L}^1$ for which $\mathbb{E}\phi(h(\omega) - \hat{\pi})$ is known exactly, then for all $\pi \in \mathcal{L}^1$ such that $\pi \geq \hat{\pi}$, we can set $U_\pi = \min\{U_\pi, \mathbb{E}\phi(h(\omega) - \hat{\pi})\}$. See Section 9.1 for additional details.

8.3 Obtaining Lower Bounds μ^k

We base our lower bounding technique on the fact that a valid lower bound can be obtained from a feasible solution to subproblem f^k. For any $\pi \in \mathcal{R}_1^k$, the overall objective value is $\bar{\psi}(\pi) + \mathbb{E}\phi(h(\omega) - \pi)$. Because this exact value is obtained from a feasible solution to subproblem f^k, this implies that it is a valid local lower bound. Moreover, μ^k defined in (13) is an improved lower bound.

For any $\mathcal{H}^k$, $\bar{\mu}^k \leq \mu^k \leq f^k$.

Proof. Observe that:

$$\begin{aligned}
\bar{\mu}^k = \psi(l^k) + \mathbb{E}\phi(h(\omega) - u^k) \leq \min_{\pi \in \{\mathcal{L}^1 \cap \mathcal{H}^k\}} \bar{\psi}(\pi) + \min_{\pi \in \mathcal{R}_1^k} \mathbb{E}\phi(h(\omega) - \pi) \leq \\
\min_{\pi \in \mathcal{R}_1^k} \bar{\psi}(\pi) + \min_{\pi \in \mathcal{R}_1^k} \mathbb{E}\phi(h(\omega) - \pi) \leq \min_{\pi \in \mathcal{R}_1^k} \left\{\bar{\psi}(\pi) + \mathbb{E}\phi(h(\omega) - \pi)\right\} \leq \\
\max_{\pi \in \mathcal{R}_1^k} \left\{\bar{\psi}(\pi) + \mathbb{E}\phi(h(\omega) - \pi)\right\} = \mu^k \leq \max_{\pi \in \{\mathcal{L}^1 \cap \mathcal{H}^k\}} \left\{\bar{\psi}(\pi) + \mathbb{E}\phi(h(\omega) - \pi)\right\} = f^k.
\end{aligned}$$

The first inequality is by Proposition (c); the second and fifth are because $\mathcal{R}_1^k \subseteq \{\mathcal{L}^1 \cap \mathcal{H}^k\}$.
■

8.4 Obtaining Upper Bounds ν^k

We use one of three approaches presented and depicted in Section 4.3 to upper bound f^k: *First-Stage Interior Bounding*, *First-Stage Exterior Bounding*, and *Second-Stage Exterior Bounding.* As in the overall objective, each upper bound consists of a first-stage bounding component and a component that bounds the second-stage expectation $\mathbb{E}\phi$. We next describe the three approaches in greater detail and demonstrate their validity in Proposition 8.4. For completeness we recall some of the definitions provided in the main body of the paper.

First-Stage Interior Bounding (Variant I1). This approach upper bounds f^k as in (14). The key idea is to search for $\pi \in \{\mathcal{L}^1 \cap \mathcal{H}^k\}$ that maximizes the sum of $\bar{\psi}(\pi)$ plus the upper bound U_π on $\mathbb{E}\bar{\phi}$.

First-Stage Exterior Bounding (Variant E1). This approach upper bounds f^k as in (15). The first-stage bounding component of ν_{E1}^k is constructed from $\bar{\psi}(\pi)$ for $\pi \in \{\mathcal{L}^1 \cap \mathcal{H}^k\}$, while the second-stage bounding component uses the nondecreasing property of ϕ and is based upon U_π values corresponding to $\pi \in \mathcal{L}^1$ exterior to $\mathcal{H}^k$. One or more $\pi \in \mathcal{L}^1$ are identified that are near, yet below, l^k in every dimension, and from these the minimum U_π value is used to upper bound $\mathbb{E}\phi(h(\omega) - l^k)$.

Specifically, we search for $\pi \in \mathcal{L}^1$ that are nearest to l^k with respect to dimension i', yet smaller, i.e. $\pi \in \mathcal{L}^1$ such that $\pi_i \leq l_i^k$ for all dimensions i. While any choice of i' could work, we choose a dimension $i' \in_{i \in \{1,\ldots,m\}} l_i^k$. Let $\mathcal{N}_l^k$ be the set containing all such $\pi \in \mathcal{L}^1$. To

complete the expectation bounding component for ν^k we choose from $\pi \in \mathcal{N}_l^k$ the minimum value of their U_π values, or if none are available, we use a weaker upper bound $\phi(\bar{h})$.
Second-Stage Exterior Bounding (Variant E2). This approach upper bounds f^k as in (16), also using $\max\limits_{\pi \in \{\mathcal{L}^1 \cap \mathcal{H}^k\}} \bar{\psi}(\pi)$ as the first-stage bounding component. Unlike the previous bounds, however, we use $\tau \in \mathcal{L}^2$ to provide an expectation bounding component for ν^k with respect to l^k. For each scenario $\omega \in \Omega$, let $\delta^\omega = h(\omega) - l^k$ (consistent with Assumption **A5**, by the construction of $\mathbf{u}$ we assume $h(\omega) \geq l^k \ \ \forall\, \omega \in \Omega$, ensuring $\delta^\omega \geq 0$). Then using the nondecreasing property of ϕ we search over all $\tau^\omega \in \mathcal{L}^2$ such that $\tau^\omega \geq \delta^\omega$ attempting to find a surrogate monoid element τ^ω that will provide an upper bound on the value of $\phi(\delta^\omega)$; this is repeated for every scenario $\omega \in \Omega$.

Specifically, we search for $\tau^\omega \in \mathcal{L}^2$ that are nearest to δ^ω with respect to dimension i', yet larger, i.e. $\tau^\omega \in \mathcal{L}^2$ such that $\tau_i^\omega \geq \delta_i^\omega$ for all dimensions i. Over all dimensions $i = 1, \ldots, m$, we choose dimension $i' \in_{i \in \{1,\ldots,m\}} \delta_i^\omega$ (again, any choice of i' could work). Let $\mathcal{N}_{\delta^\omega}^k$ be the set containing all such $\tau^\omega \in \mathcal{L}^2$. If $\mathcal{N}_{\delta^\omega}^k \neq \emptyset$ for scenario ω, among these τ^ω we use the minimum of the corresponding $\bar{\phi}(\tau^\omega)$ to serve as an upper bound on $\phi(\delta^\omega)$. If no such τ^ω exists, we instead use the value $\phi(\bar{h})$ as an upper bound on $\phi(\delta^\omega)$. In both cases, the expectation over all such values is a valid upper bound for $\mathbb{E}\phi(\delta^\omega)$.

For any $\mathcal{H}^k$, ν_{I1}^k, ν_{E1}^k, and ν_{E2}^k are valid upper bounds on f^k.
Proof. For ν_{I1}^k we have:

$$f^k = \max_{\pi \in \{\mathcal{L}^1 \cap \mathcal{H}^k\}} \left\{\bar{\psi}(\pi) + \mathbb{E}\phi(h(\omega) - \pi)\right\} \leq \max_{\pi \in \{\mathcal{L}^1 \cap \mathcal{H}^k\}} \left\{\bar{\psi}(\pi) + U_\pi\right\} = \nu_{I1}^k,$$

where the relation follows from the definition of U_π.

For ν_{E1}^k we first note:

$$\max_{\pi \in \{\mathcal{L}^1 \cap \mathcal{H}^k\}} \mathbb{E}\phi(h(\omega) - \pi) \leq \min_{\pi \in \mathcal{N}_l^k} U_\pi \leq \phi(\bar{h}). \tag{17}$$

Applying Proposition (c) to (17) gives:

$$f^k = \max_{\pi \in \{\mathcal{L}^1 \cap \mathcal{H}^k\}} \left\{\bar{\psi}(\pi) + \mathbb{E}\phi(h(\omega) - \pi)\right\} \leq \max_{\pi \in \{\mathcal{L}^1 \cap \mathcal{H}^k\}} \bar{\psi}(\pi) + \max_{\pi \in \{\mathcal{L}^1 \cap \mathcal{H}^k\}} \mathbb{E}\phi(h(\omega) - \pi) \leq$$

$$\max_{\pi \in \{\mathcal{L}^1 \cap \mathcal{H}^k\}} \bar{\psi}(\pi) + \min\left\{\min_{\pi \in \mathcal{N}_l^k} U_\pi,\ \phi(\bar{h})\right\} = \nu_{E1}^k.$$

For ν_{E2}^k let $\delta^\omega = h(\omega) - l^k$, and note that:

$$\phi(\delta^\omega) \leq \min_{\tau^\omega \in \mathcal{N}_{\delta^\omega}^k} \phi(\tau^\omega) \leq \phi(\bar{h}). \tag{18}$$

Applying Proposition (c) to (18) gives:

$$f^k = \max_{\pi \in \{\mathcal{L}^1 \cap \mathcal{H}^k\}} \left\{\bar{\psi}(\pi) + \mathbb{E}\phi(\delta^\omega)\right\} \leq \max_{\pi \in \{\mathcal{L}^1 \cap \mathcal{H}^k\}} \bar{\psi}(\pi) + \max_{\pi \in \{\mathcal{L}^1 \cap \mathcal{H}^k\}} \mathbb{E}\phi(\delta^\omega) \leq$$

$$\max_{\pi \in \{\mathcal{L}^1 \cap \mathcal{H}^k\}} \bar{\psi}(\pi) + \mathbb{E}\min\left\{\min_{\tau^\omega \in \mathcal{N}_{\delta^\omega}^k} \phi(\tau^\omega),\ \phi(\bar{h})\right\} = \nu_{E2}^k.$$

$\nu_{I1}^k \le \nu_{E1}^k$. ∎

Proof. Note that:

$$\nu_{I1}^k = \max_{\pi \in \{\mathcal{L}^1 \cap \mathcal{H}^k\}} \left\{\bar{\psi}(\pi) + U_\pi\right\} \le \max_{\pi \in \{\mathcal{L}^1 \cap \mathcal{H}^k\}} \bar{\psi}(\pi) + \max_{\pi \in \{\mathcal{L}^1 \cap \mathcal{H}^k\}} U_\pi \le \tag{19}$$

$$\max_{\pi \in \{\mathcal{L}^1 \cap \mathcal{H}^k\}} \bar{\psi}(\pi) + \min\left\{\min_{\pi \in \mathcal{N}_l^k} U_\pi, \phi(\bar{h})\right\} = \nu_{E1}^k. \tag{20}$$

The last inequality is due to the nonincreasing nature of U_π with respect to π. ∎

In contrasting the expectation bounding components of ν_{I1}^k, ν_{E1}^k and ν_{E2}^k, both ν_{I1}^k and ν_{E1}^k use $\bar{\psi}$, while ν_{E2}^k uses $\bar{\phi}$. Whereas ν_{E1}^k is designed to quickly identify an expectation bounding component based on $\pi \in \mathcal{L}^1$ nearest to a certain dimension i', its bound is weaker than that of ν_{I1}^k (see Proposition 8.4). However, as compared to ν_{E1}^k, the interior of $\mathcal{H}^k$ is a potentially larger space for ν_{I1}^k over which to search, and performance trade-offs likely exist. While ν_{E2}^k is also built to quickly identify an upper bound based on a single dimension i', we must repeat the process for all scenarios $\omega \in \Omega$ to generate an expectation bounding component composed of $|\Omega|$ individual bounds.

9 Search Space Reduction

This section details our two-fold approach to reduce the search space, with the first eliminating suboptimal π from $\mathcal{L}^1$, and the second eliminating entire subregions of $\mathcal{H}^k$ (or $\mathcal{H}^k$ itself).

9.1 Removal of Suboptimal Monoid Elements

Proposition 8.2 enables us to remove suboptimal $\pi \in \mathcal{L}^1$ from consideration and thereby avoid computing $\mathbb{E}\phi(h(\omega) - \pi)$ for such π. With respect to a reference $\hat{\pi} \in \mathcal{L}^1$ for which $\mathbb{E}\phi(h(\omega) - \hat{\pi})$ is exactly known, it provides, without explicit calculation, a valid upper bound on the expectation for any $\pi \in \mathcal{L}^1$ such that $\pi \ge \hat{\pi}$. It uses global lower bound L as a criterion to cut off inferior $\pi \in \mathcal{L}^1$, as it is only necessary to retain those $\pi \in \mathcal{L}^1$ for which $\bar{\psi}(\pi) + \mathbb{E}\phi(h(\omega) - \pi) \ge L$. That is, if for any $\pi \in \mathcal{L}^1$ we have $\bar{\psi}(\pi) + U_\pi < L$, we immediately remove π from $\mathcal{L}^1$, simultaneously bypassing the need for costly expectation computation for π. We use it in **Line 18** of **GBB**, after a given $\mathcal{H}^k$ has been considered, to remove $\pi \in \mathcal{L}^1$ that are shown to be suboptimal. Note that we attempt to update L in both **Line 11** of **GBB** and **Line 11** of Procedure 3.

9.2 Reducing and Fathoming $\mathcal{H}^k$

We fathom $\mathcal{H}^k$ whenever $\{\mathcal{L}^1 \cap \mathcal{H}^k\} = \emptyset$ (**Lines 3 – 4** of Procedure 3). Moreover, if $|\mathcal{L}^1 \cap \mathcal{H}^k| = 1$ (i.e., there is exactly one $\pi \in \{\mathcal{L}^1 \cap \mathcal{H}^k\}$), we immediately set $l^k = u^k = \pi$, a singleton (**Lines 13 – 15** of Procedure 3). **Lines 8 – 11** of **GBB** will subsequently handle such a reduced hyper-rectangle, where it will either be fathomed or, if $L < \mu^k$, stored as the new incumbent π^*.